Fully-Dualizable and Invertible $\mathcal{E}_n$-Algebras

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Abstract

We prove a conjecture of Lurie, Brochier, Jordan, Safronov and Snyder [Lur09b, BJSS21] characterizing fully-dualizable and invertible $\mathcal{E}_n$-algebras seen as objects in the higher Morita category $\mathbf{Mor}_n(\mathcal{V})$ [Lur09b, Sch14, Hau17b, Hau23]. In other words, we characterize those $\mathcal{E}_n$-algebras which give rise to $(n+1)$-dimensional topological quantum field theories (TQFT), and those which give rise to an invertible one.

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1 Introduction

One of the central insights in the modern study of topological quantum field theory (TQFT) is that the structure of a field theory is governed by higher-categorical dualizability conditions. This idea originates in the pioneering work of Baez-Dolan [BD95], who formulated the Cobordism Hypothesis. They propose that an n-dimensional fully extended TQFT should be the data of: a symmetric monoidal (∞, m)-category $\mathcal{A}$, and a "fully-n-dualizable" object $a \in \mathcal{A}$. The initial such TQFT is then conjectured to be given by the (∞, n)-category of framed bordisms. Typically, the symmetric monoidal (∞, m)-categories $\mathcal{A}$ of interest arise from studying algebraic structures and the Cobordism Hypothesis can be seen as a duality between algebra and manifold topology.

Lurie observed in [Lur09b, Sec. 4.1] that the collection of $\mathcal{E}_n$-algebras, in some fixed symmetric monoidal $(\infty, 1)$-category $\mathcal{V}$ with compatible colimits, can be arranged into such a symmetric monoidal $(\infty, n+1)$-category $\mathbf{Mor}_n(\mathcal{V})$ called the higher Morita category. This generalizes the well-known classical Morita $(2, 2)$-category whose objects are non-commutative rings and whose morphisms are bimodules. The idea of this category was first sketched by Lurie in a [Lur09b, Def. 4.1.11] and later formalized by Scheimbauer and Haugseng in [Sch14, Hau17a, Hau23]. It follows from the construction that all k-morphisms in $\mathbf{Mor}_n(\mathcal{V})$, for $1 \leq k < n$, have both left and right adjoints, implying that any $\mathcal{E}_n$-algebras A automatically gives rise to a n-dimensional fully-extended TQFT. The computations of the partition function of this TQFT is exactly given by factorization homology with coefficients in A as predicted by Lurie [Lur09b, Thm. 4.1.24] and formalized in [Sch14]. The n-morphisms of $\mathbf{Mor}_n(\mathcal{V})$ do not however necessarily have adjoints, which invites the question of characterizing those $\mathcal{E}_n$-algebras A for which the associated TQFT extends to $(n+1)$-dimensions, i.e., the fully-dualizable $\mathcal{E}_n$-algebras. The purpose of this paper is to answer this question, and further classify which $\mathcal{E}_n$-algebras form an invertible TQFT.

Classically, the fully-dualizable objects of the classical Morita $(2, 2)$-category are known as smooth and proper algebras, and the invertible objects are known as Azumaya algebras. These notions were generalized to the dg-setting by Toën [Toë12]. More explicitly, an algebra A in ${}_k\mathbf{Mod}$ is smooth if A is perfect as an $A^{op} \otimes_k A$-module, and proper if A is perfect as a k-module. A smooth and proper algebra A is further Azumaya if the canonical

$$k \xrightarrow{\sim} \mathbf{HC}_{\mathcal{E}_1}(A)$$
$$A^{\mathrm{op}} \otimes_k A \xrightarrow{\sim} \mathbf{End}_k(A)$$

are equivalences.

Lurie claimed in [Lur09b, Rem. 4.1.27] that unwinding the proof of the Cobordism Hypothesis should give an explicit generalization of the above conditions for full-dualizability of $\mathcal{E}_n$-algebras, involving factorization homology. This criterion was later stated as a conjecture by Brochier, Jordan, Safronov, Snyder [BJSS21, Conj. 1.8], which we prove in more generality in Theorem A stated below using the language of factorization homology [AFT17b, AFR18b, AFT17a].

Theorem 1.1 (Thm. A)

Let $0 \leq k < n$ and $\mathcal{V} \in \mathbf{Alg}_{\mathcal{E}_\infty}(\mathbf{Cat}_{(\infty,1)})$ whose tensor product commutes with sifted colimits in both variables. Let M be a k-morphism in $\mathbf{Mor}_n(\mathcal{V})$. Then M is fully-dualizable if and only if, for all $0 \leq i \leq n-k$, the canonical $\mathcal{E}_1$-actions

$$\int_{(S^{i-1},D^i)}(M, A) \curvearrowright M \curvearrowleft \int_{(S^{i-1},D^i)}(M, B)$$

both exhibit M as a dualizable module, where A (resp. B) are either the (co)domain of M if $1 \leq k$, or the identity $\mathbb{1}_{\mathcal{V}}$ if $k = 0$.

One can interpret the factorizations $\int_{(S^{i-1},D^i)}(M, A)$ as a higher version of Hochschild Homology of M relative to A. Brochier *et al.* also conjectured a generalization of the characterization of Azumaya algebras [BJSS21, Conj. 1.9], which we prove in in Theorem B stated below.

Theorem 1.2 (Thm. B)

Let $0 \leq k < n$ and $\mathcal{V} \in \mathbf{Alg}_{\mathcal{E}_\infty}(\mathbf{Cat}_{(\infty,1)}))$. Let M be a k-morphism in $\mathbf{Mor}_n(\mathcal{V})$. Then M is invertible if and only if it is fully-dualizable and, for all $1 \leq i \leq n-k$, the universal $\mathcal{E}_{n-k-i+1}$-algebra maps

$$\int_{(S^{i-1},D^i)}(M, A) \xrightarrow{\sim} \mathbf{HC}^B_{\mathcal{E}_{n-k-i}}(M)$$
$$\int_{(S^{i-1},D^i)}(M, B) \xrightarrow{\sim} \mathbf{HC}^A_{\mathcal{E}_{n-k-i}}(M)$$

are invertible, where A (resp. B) are either the (co)domain of M if $1 \leq k$, or the identity $\mathbb{1}_{\mathcal{V}}$ if $k = 0$.

1.1 Overview

Section 2 To establish these results, we first need to prove a few preliminary results about (∞, n)-categories with adjoints in section 2. Most of these results are well-known to experts, but lacking of any rigorous proof in the literature, so we include them here for completeness. All of the work of this section heavily relies on the seminal result of Riehl and Verity [RV16, Thm. 4.4.11] bringing adjunctions in $(\infty, 2)$-categories into a powerful homotopy coherent framework. One of the most important of these folklore theorem is that the inclusion $\mathbf{Cat}^{\mathrm{adj}}_{(\infty,n)} \hookrightarrow \mathbf{Cat}_{(\infty,n)}$ has left and right adjoints (see Proposition 2.11). The existence of the left adjoint is trivial from the definition, but the existence of the right adjoint is more subtle. One can construct it explicitly by hand, realizing the idea that it ought to be given by the full sub-(∞, n)-category on these morphisms which are fully-adjointable. We choose here to construct this functor more formally, using the Adjoint Functor Theorem [Lur18, Thm. 06Q5] by proving that $\mathbf{Cat}^{\mathrm{adj}}_{(\infty,n)} \hookrightarrow \mathbf{Cat}_{(\infty,n)}$ preserves colimits:

Proposition 1.3 (Proposition 2.8)

Let $F \colon I \to \mathbf{Cat}_{(\infty,n)}$ be a diagram of (∞, n)-categories which all admit right (resp. left) adjoints for all k-morphisms. Then so does $\varinjlim F$.

The argument relies on the explicit way in which colimits of (∞, n)-categories are constructed. More explicitly, a k-morphism in some $\varinjlim_{i \in \mathcal{I}} \mathcal{C}_i$ is always given by some intricate pasting of k-morphisms in the $\mathcal{C}_i$'s. This is an intrinsic consequence of the "Segalification" construction for (∞, n)-categories described in [BS24]. It is then well-known that composition of adjoints should have adjoints, which will in turn imply that $\mathbf{Cat}^{\mathrm{adj}}_{(\infty,n)}$ is stable under colimits. The main advantage of this approach is that it gives a general method for proving that some category, defined as a colimit, has adjoints. The author sees this as a powerful tool to show results of the type: "to construct a fully-adjointable (resp. dualizable) morphism (resp. object), one only needs to exhibit this and that (co)unit". A first statement of the type is given by the result bellow:

Proposition 1.4 (Proposition 2.15)

The free $(\infty, k{+}1)$-category on a single fully-adjointable k-morphism $\mathsf{Free}^{\{k\}}(c_k)$ is the colimit of the diagram

$$\begin{tikzcd}
 & & & \mathsf{Free}^{\{k\}}(c_k) & & & \\
\cdots \arrow[urrr, dashed, two heads] & \mathbf{Adj}_k \arrow[urr, dashed, two heads] & & \mathbf{Adj}_k \arrow[u, dashed, two heads] & & \mathbf{Adj}_k \arrow[ull, dashed, two heads] & \cdots \arrow[ulll, dashed, two heads] \\
c_k \arrow[ul, two heads, "R"] \arrow[ur, two heads, "L"'] & c_k \arrow[ul, two heads, "R"] \arrow[ur, two heads, "L"'] & & c_k \arrow[ul, two heads, "R"] \arrow[ur, two heads, "L"'] & & c_k \arrow[ul, two heads, "R"] \arrow[ur, two heads, "L"'] &
\end{tikzcd}$$

where all projections are epimorphisms, and $\mathbf{Adj}_k = \sigma^{k-1}(\mathbf{Adj})$ is the walking adjoint k-morphism.

In essence, this can be seen as formalizing the idea: "to give a fully-adjoint 1-morphism is equivalent to providing a sequence of candidates for the iterated adjoints and for the (co)units". The proof of the main Theorem relies on a similar guiding principle: "to show f is fully-adjoint, it suffices to find adjoints for the iterated (co)units of the form $\eta(\epsilon(\epsilon(\ldots \eta(f) \ldots)))$" (where we range over all words in ϵ and η of some fixed length).

Section 3 In the following section, we then focus on the geometric and algebraic tools required for our proofs. On the geometric side, the main ingredient here is the powerful theory of factorization homology of conically smooth stratified spaces developed by Ayala, Francis, Rozenblyum and Tanaka in [AFT17b, AFR18b, AFT17a]. For us, factorization homology is the natural tool to manipulate these $\mathcal{E}_k$-algebras in iterated bimodules M. Following the guiding principle above, factorization homology will allow us to explicitly describe the algeraic structure corresponding to each of these iterated (co)units $\epsilon(\eta(\epsilon(\ldots \eta(M) \ldots)))$. We define various stratified spaces which will "carry" this dualizability data, and various stratified map which induces functors sending "M" to these "$\eta(\eta(\epsilon(\ldots \epsilon(M) \ldots)))$". These spaces can be explicitly visualized as stratifications of the usual handlebodies. Their algebraic structure can be visualized as iteratively "pinching" these handlebodies, through constructible bundle maps.

Finally, we prove a couple technical algebraic results. One of the main ingredient in our proof of Theorem A is a criterion (Proposition 3.11) for when n-morphisms in $\mathbf{Mor}_n(\mathcal{V})$ have adjoints, essentially generalizing

[Lur09b, Prop. 4.6.2.13]. Finally, to complete our proof of Theorem B, we also need to understand how Hochschild cohomology appears as duality data in $\mathbf{Mor}_n(\mathcal{V})$, which is done in Proposition 3.15.

Section 4 This short section is the compiled proofs of both our main Theorems.

1.2 Acknowledgments

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1.3 Notations

- For $n \geq 0$, we denote by $\mathbf{Cat}_{(\infty,n)}$ the presentable $(\infty,1)$-category of (∞,n)-categories. We have a sequence of reflections and coreflections:

$$\begin{tikzcd}
\mathcal{S} \arrow[r, hook] & \mathbf{Cat}_{(\infty,1)} \arrow[l, bend right, "|-|_{>0}"'] \arrow[l, bend left, "-_{\leq 0}"] \arrow[r, hook] & \mathbf{Cat}_{(\infty,2)} \arrow[l, bend right, "|-|_{>1}"'] \arrow[l, bend left, "-_{\leq 1}"] \arrow[r, hook] & \cdots \arrow[l, bend right, "|-|_{>2}"'] \arrow[l, bend left, "-_{\leq 2}"] \arrow[r, hook] & \mathbf{Cat}_{(\infty,n)} \arrow[l, bend right, "|-|_{>n-1}"'] \arrow[l, bend left, "-_{\leq n-1}"] \arrow[r, hook] & \cdots \arrow[l, bend right, "|-|_{>n}"'] \arrow[l, bend left, "-_{\leq 0}"]
\end{tikzcd}$$

- For $n \geq 0$, we have an adjunction

$$\begin{tikzcd}
\mathbf{Cat}_{(n,n)} \arrow[r, hook] & \mathbf{Cat}_{(\infty,n)} \arrow[l, bend right, "\mathcal{H}o_n"']
\end{tikzcd}$$

- We denote by c_k the walking k-morphism, which we can see as an (∞,n)-category for all $n \geq k$. We denote by ∂c_k the boundary of the walking k-morphism, i.e., the walking pair of parallel $(k-1)$-morphisms, which can also be seen as an (∞,n)-category for all $n \geq k-1$. It is the colimit of the diagram

$$\begin{tikzcd}
c_0 \arrow[r] \arrow[dr] & c_1 \arrow[r] \arrow[dr] & \cdots \arrow[r] \arrow[dr] & c_{k-1} \arrow[dr, dashed] & \\
 & & & & \partial c_k \\
c_0 \arrow[r] \arrow[ur] \arrow[urrr, dashed] & c_1 \arrow[r] \arrow[ur] \arrow[urr, dashed] & \cdots \arrow[r] \arrow[ur] \arrow[ur, dashed] & c_{k-1} \arrow[ur, dashed] &
\end{tikzcd}$$

  A k-morphism $f\colon c_k \to \mathcal{C}$ is also denoted $f\colon a_{k-1} \to b_{k-1} \colon \ldots \colon a_0 \to b_0 \colon \mathcal{C}$ where $a_i, b_i \colon c_i \to \mathcal{C}$ are the induced projections above.

- We let $\mathbf{Cat}_{(\infty,1)}^{\mathsf{geom}-\mathrm{colim}}$ denote the symmetric monoidal $(\infty,1)$-category whose:
  (a) **objects** are $(\infty,1)$-categories admitting geometric realizations.
  (b) **morphisms** are functors preserving geometric realizations.

  The symmetric monoidal structure is given by [Lur09a, Cor. 4.8.1.4].

- A refinement of stratified space will always be denoted by a dashed arrow $X \dashrightarrow Y$. A (weak) constructible bundle will always be denoted by a double-headed arrow $X \twoheadrightarrow Y$. An open embedding will always be denoted by a hooked arrow $X \hookrightarrow Y$ or by $X \xrightarrow{\sim} Y$ if it is an isotopy equivalence. A closed inclusion will always be denoted by a hooked arrow $X \overset{\mathrm{cl}}{\hookrightarrow} Y$ with the label "cl".

- We let $S^{-1} = \emptyset$ and $D^0 = \{\bullet\}$.

- Throughout most of the paper, we fix a symmetric monoidal $(\infty,1)$-category $\mathcal{V}$ which has sifted colimits and whose tensor product commutes with sifted colimits in both variables.

- We denote operads by $\underline{\mathcal{O}} \to \mathbf{FinSet}^*$ with an underline. Their "underlying category", *i.e.* the fiber $\underline{\mathcal{O}}^{\{\bullet\}^+}$ over $\{\bullet\}^+$, is denoted by $\mathcal{O}$ without an underline.

2 Adjoints in (∞, n)-categories

Throughout this section, we establish a few "folklore" results about (∞, n)-categories with adjoints, which are crucial for the proof of the main Theorems. However expected these results may seem, we could not find any rigorous proof of them in the literature, so we include them here for completeness. The reader can safely skip this section on a first reading, and come back to it when needed.

For the interested reader, however, one can summerize our approach as follows. To work with robust homotopically-coherent notion, we define fully-adjointable morphism in a $(\infty, n+1)$-category $\mathcal{C}$ has the morphisms that appear in the image of $\mathsf{Sub}^{\{1,\dots,n\}}(\mathcal{C}) \hookrightarrow \mathcal{C}$, where $\mathsf{Sub}^{\{1,\dots,n\}}(\mathcal{C})$ is the full sub-$(\infty, n+1)$-category with adjoints. To make sense of this definition, we then need to show that $\mathsf{Sub}^{\{1,\dots,n\}}(\mathcal{C})$ is well-defined, and that the functor $\mathsf{Sub}^{\{1,\dots,n\}}(\mathcal{C}) \hookrightarrow \mathcal{C}$ is a monomorphism (see Proposition 2.11), which is the content of the second subsection 2.2. One way to tackle this problem is to show that colimits of (∞, n)-categories with adjoints have adjoints (see Proposition 2.8), which is the content of the first subsection 2.1. The advantage of this strategy is that it pushes us to develop tools to decide when a colimit of (∞, n)-categories has adjoints in general. This becomes important to link our definition of fully-adjointable morphisms to the more classical one: has an adjoint, with (co)units having adjoints, with (co)units having adjoints, and so on. This is the content of Corollary 2.16, the main result of subsection 2.3. Finally, we finish this section by proving that the classifying space of the free/walking full-adjunction is contractible, which is the content of subsection 2.4 and will be useful for our second criterion, the Theorem 4.5

Before heading into the content of this section, we first state the definition of an adjoint morphism and the celebrated Theorem of Riehl and Verity [RV16, Thm. 4.4.11].

Definition 2.1 (Adjoint)**.** Let $n > k \geq 1$ and $\mathcal{C} \colon \mathbf{Cat}_{(\infty,n)}$. We say that $l \colon a_{k-1} \to b_{k-1} \colon \dots \colon a_0 \to b_0 \colon \mathcal{C}$ is a left adjoint if its equivalence class in $\mathcal{H}o_2(\mathcal{C}(a_0, b_0)(a_1, b_1) \dots (a_{k-2}, b_{k-2}))$ is a left adjoint.

Construction 2.1 (Free/Walking Adjunction)*.* Recall that Schanuel and Street [SS86] defined combinatorially the free $(2,2)$-category $\mathbf{Adj}$ containing an adjunction $L \colon + \underset{\longleftarrow}{\overset{\longrightarrow}{\perp}} - : R$. This category was later studied in depth by Riehl and Verity [RV16], in a higher categorical context.

The results of this paper rely greatly on the powerful Theorem bellow.

Theorem 2.1 (Homotopy Uniqueness of Adjunctions [RV16, Thm. 4.4.11])

Let $\mathcal{C} \colon \mathbf{Cat}_{(\infty,2)}$. Precomposition by $L, R \colon c_k \to \mathbf{Adj}$ gives monomorphisms

$$[\mathbf{Adj}, \mathcal{C}] \underset{R^*}{\overset{L^*}{\rightrightarrows}} [c_1, \mathcal{C}]$$

selecting the connected components of left (resp. right) adjoint morphisms.

2.1 Colimits of (∞, n)-categories with adjoints have adjoints

In this section, we prove that a colimit of a diagram of (∞, n)-categories are given as compositions of k-cells in the categories in the diagram. For this purpose, we use the following equivalent definitions of of (∞, n)-categories as:

- Rezk's complete Segal Θ_n-spaces [Rez10], inspired by Joyal's unpublished ideas [Joy97],
- Barwick's n-fold complete Segal spaces [Bar05], inspired by Rezk's complete segal spaces (for $n = 1$) [Rez01],

In [BSP20], Barwick and Schommer-Pries proved that these two definitions are equivalent. In [Hau17b], Haugseng further proved that the two theories coincide priori to univalent completion. Combining both results, we get the diagram where left (resp. right) adjoints commute with the vertical equivalences, and where the last column is equivalent to $\mathbf{Cat}_{(\infty,n)}$.

$$\begin{tikzcd}
\mathcal{P}\mathrm{sh}(\Delta^n) \arrow[r, bend left, "\mathbb{L}_S"] \arrow[r, phantom, "\perp"] & \mathbf{Seg}(\Delta^n) \arrow[l, bend left] \arrow[r, bend left, "\mathbb{L}_C^{\Delta}"] \arrow[r, phantom, "\perp"] \arrow[d, "\wr"] & \mathbf{Seg}^{\mathrm{univ}}(\Delta^n) \arrow[l, bend left] \arrow[d, "\wr"] \\
& \mathbf{Seg}(\Theta^n) \arrow[r, bend left, "\mathbb{L}_C^{\Theta}"] \arrow[r, phantom, "\perp"] & \mathbf{Seg}^{\mathrm{univ}}(\Theta^n) \arrow[l, bend left]
\end{tikzcd} \tag{2.1}$$

Note that an explicit formula for the Δ^n-Segalification $\mathbb{L}_S^{\Delta}$ was established in [BS24, Thm. D] by Barkan and Steinebrunner following the theory of necklaces developped by Dugger and Spivak [DS11]. Another formula was

established independently for $n = 1$ by Ayala and Francis [AF24, Sec. 3]. An explicit formula for Θ^n-univalent completion $\mathbb{L}_C^{\Theta}$ was established in [AF18, Prop. 2.4], generalizing the Rezk's original formula [Rez01]. Using these formulas and using that left adjoints preserves colimits, we can understand explicitly how the k-cells of a colimit of categories are built from the k-cells of the categories in the diagram.

Construction 2.2 (Suspension). Recall that we have a functor $\sigma\colon \mathcal{P}\mathrm{sh}(\Delta^n) \to \mathcal{P}\mathrm{sh}(\Delta^{n+1})$ defined as follows:

$$\sigma X_{a_1,\ldots,a_{n+1}} = \{\bullet_-\} \sqcup \bigsqcup_{i=1}^{a_1} X_{a_2,\ldots,a_{n+1}} \sqcup \{\bullet_+\}$$

where the maps are defined in the obvious way. Note that if X is reduced/Segal (in the ith direction)/univalent, then σX is reduced/Segal (in the $(i+1)^{\text{th}}$ direction)/univalent. We can iterate this construction and get

$$\sigma^k X_{a_1,\ldots,a_{n+k}} = (\partial c_k)_{a_1,\ldots,a_{k-1},0,\ldots,0} \bigsqcup \left(X_{a_{k+1},\ldots,a_{n+k}} \times \prod_{i=1}^{k} [a_i] \right)$$

Note that $\sigma^k c_l = c_{l+k}$. We also have an adjunction

$$\begin{tikzcd} \mathbf{Cat}_{(\infty,n)} \arrow[r, bend right, "\sigma^k"'] \arrow[r, phantom, "\top"] & \mathbf{Cat}_{(\infty,n+k)} \arrow[l, bend right, "\partial c_k \backslash\ \hom"'] \end{tikzcd}$$

where for $a_{k-1}, b_{k-1}\colon \ldots : a_0 \to b_0 : \mathcal{C}$, we denote the corresponding $\hom(\partial c_k \to \mathcal{C}) = \mathcal{C}(a_0, b_0)(a_1, b_1)\ldots(a_{k-2}, b_{k-2})$.

We denote by $\mathbf{Adj}_k = \sigma^{k-1}\mathbf{Adj}$ the free $(\infty, k+1)$-category containing an adjoint k-morphism. We have an immediate corollary of Theorem 2.1.

Corollary 2.2 (Homotopy Uniqueness of (higher) Adjunctions [RV16, Thm. 4.4.11])

Let $1 \le k < n$ and $\mathcal{C}\colon \mathbf{Cat}_{(\infty,n)}$. Precomposition by $L, R\colon c_k \to \mathbf{Adj}_k$ gives monomorphisms

$$\begin{tikzcd} {[\mathbf{Adj}_k, \mathcal{C}]} \arrow[r, hook, bend left, "L^*"] \arrow[r, hook, bend right, "R^*"'] & {[c_k, \mathcal{C}]} \end{tikzcd}$$

selecting the connected components of left (resp. right) adjoint k-morphisms.

Proof: σ^{k-1} is a left adjoint composed with a forgetful from a slice, so it preserves contractible colimits and thus epimorphisms. Additionally, the notion of adjoint k-morphism is a suspension of that of adjoint 1-morphisms. ■

To formulate the next few lemmas, we now extend the notion of "having adjoints" to n-fold simplicial spaces. Note that the homotopy uniqueness results above do not apply to general n-fold simplicial spaces, only to Segal and univalent/complete ones. In other words, $c_k \to \mathbf{Adj}_k$ is not an epimorphism in $\mathcal{P}\mathrm{sh}(\Delta^n)$ or $\mathbf{Seg}(\Delta^n)$ for $k < n$.

Definition 2.2 (n-Fold Simplicial Space with Adjoints)**.** Let $1 \le k < n$. Let $X \in \mathcal{P}\mathrm{sh}(\Delta^n)$ be a reduced n-fold simplicial space. We say that X has right (resp. left) adjoints for all k-cells if for all $c_k \to X$ for which there exists a (not necessarily unique) lift

$$\begin{tikzcd} c_k \arrow[r] \arrow[d] & X \\ \mathbf{Adj}_k \arrow[ur, dashed] & \end{tikzcd}$$

where the downward functor is L (resp. R).

In the next three results, we describe how Segalification interacts with having adjoints. The first lemma bellow, and its immidiate corrolary, are the key to the inductive argument in Lemma 2.5. Recall from [BS24, Thm. D] that, for X reduced, the Segalification functor can be computed as $\mathbb{L}_S^{\Delta} = \mathbb{L}_{S,n}^{\Delta} \circ \ldots \circ \mathbb{L}_{S,1}^{\Delta}$.

Lemma 2.3

Let $1 \le i$ and X be a n-fold Segal space. Then $\mathbb{L}_{S,i}^{\Delta}\left(\sigma^i X \bigsqcup_{c_{i-1}} \sigma^i X\right)$ is an $(n+i+1)$-fold Segal space.

Proof: It suffices to prove the result for $i = 1$ as Segalification is seen to commute with suspension. By Remark 2.2, we have

$$\left(\sigma X \bigsqcup_{c_0} \sigma X\right)_{a_1,\ldots,a_{n+1}} = \{\bullet_0\} \sqcup \bigsqcup_{i=1}^{a_1} X_{a_2,\ldots,a_{n+1}} \sqcup \{\bullet_1\} \sqcup \bigsqcup_{i=1}^{a_1} X_{a_2,\ldots,a_{n+1}} \sqcup \{\bullet_2\}$$

By inspection, using the Necklace formula [BS24, Thm. A], we get

$$\left(\mathbb{L}^{\Delta}_{S,i}\left(\sigma X \bigsqcup_{c_0} \sigma X\right)\right)_{a_1,\ldots,a_n} = \{\bullet_0\} \sqcup \bigsqcup_{i=1}^{a_1} X_{a_2,\ldots,a_{n+1}} \sqcup \{\bullet_1\} \sqcup \bigsqcup_{i=1}^{a_1} X_{a_2,\ldots,a_{n+1}} \sqcup \{\bullet_2\} \bigsqcup_{1\leq i<j\leq a_1} X^2_{a_2,\ldots,a_{n+1}}$$

This is easily seen to be Segal. ■

Corollary 2.4 (Composition of Adjoints)

Let $1 \leq i < k < n$. There are lifts

$$\begin{tikzcd} \mathbf{Adj}^k \arrow[d, dashed] & c_k \arrow[l, "L"'] \\ \mathbb{L}^{\Delta}_{S,i}\left(\mathbf{Adj}^k \bigsqcup_{c_{i-1}} \mathbf{Adj}^k\right) & c_k \bigsqcup_{c_{i-1}} c_k \arrow[l] \arrow[u] \end{tikzcd} \qquad \begin{tikzcd} \mathbf{Adj}^k \arrow[d, dashed] & c_k \arrow[l, "L"'] \\ \mathbb{L}^{\Delta}_{S,k+1} \circ \mathbb{L}^{\Delta}_{S,k}\left(\mathbf{Adj}^k \bigsqcup_{c_{k-1}} \mathbf{Adj}^k\right) & c_k \bigsqcup_{c_{k-1}} c_k \arrow[l] \arrow[u] \end{tikzcd}$$

Proof: By Lemma 2.3 above, $\mathbb{L}^{\Delta}_{S,i}\left(\mathbf{Adj}^k \bigsqcup_{c_{i-1}} \mathbf{Adj}^k\right)$ is Segal. Note that $\mathbb{L}^{\Delta}_{S,k+1} \circ \mathbb{L}^{\Delta}_{S,k}\left(\mathbf{Adj}^k \bigsqcup_{c_{k-1}} \mathbf{Adj}^k\right)$ is also automatically Segal by Remark 2.2. The only equivalences in these n-fold simplicial sets are the identities, hence they are also complete. By Corollary 2.2, it then suffices to show in classical (strict) (n, n)-category theory that an ith-composition of k-adjoints is an adjoint. This is a standard result from ordinary category theory. ■

Using the two results above, we can now prove this key technical lemma bellow to handle adjoints in the Segalification of a reduced n-fold simplicial space.

Lemma 2.5 (Segalification Preserves Adjoints)

Let X be a reduced n-fold simplicial space that has right (resp. left) adjoints for all k-cells. Then its Segalification $\mathbb{L}^{\Delta}_{S}(X)$ has right (resp. left) adjoints for all k-cells.

Proof: Using $\mathbb{L}^{\Delta}_{S} = \mathbb{L}^{\Delta}_{S,n} \circ \ldots \circ \mathbb{L}^{\Delta}_{S,1}$ for X reduced, and using [BS24, Lem. 2.15.a] which ensures X remains reduced through these steps, one can observe that it is sufficient to prove the result separately: for $\mathbb{L}^{\Delta}_{S,i}$ with $1 \leq i \leq k-1$, and for $\mathbb{L}^{\Delta}_{S,k+1} \circ \mathbb{L}^{\Delta}_{S,k}$.

- **Case $i < k$:** Fix $c_k = \Delta_{1,\ldots,1,0,\ldots,0} \to \mathbb{L}^{\Delta}_{S,i}(X)$. By the Necklace formula [BS24, Thm. A] and by reducedness, we can find $m \geq 1$ and a diagram

$$\begin{tikzcd} c_k \arrow[r] & \Delta_{1,\ldots,1,m,1,\ldots,1,0,\ldots,0} \arrow[r] & \mathbb{L}^{\Delta}_{S,i}(X) \\ & c_k \bigsqcup_{c_{i-1}} \ldots \bigsqcup_{c_{i-1}} c_k \arrow[r] \arrow[u, "\simeq_i"'] & X \arrow[u] \end{tikzcd}$$

where the arrows marked by $\simeq_i$ are Segal equivalences for the ith-coordinate Segal conditions. Without loss of generality, it suffices to consider the case $m = 2$. Using the universal property of $\mathbb{L}_{S,i}$ and using Corollary 2.4, we get a diagram

$$\begin{tikzcd} c_k \arrow[rr, "L"] \arrow[d] & & \mathbf{Adj}_k \arrow[dl, "2.4"'] \arrow[dd, dashed] \\ \Delta_{1,\ldots,1,2,1,\ldots,1,0,\ldots,0} \arrow[r] \arrow[ddr] & \mathbb{L}_{S,i}\left(\mathbf{Adj}_k \bigsqcup_{c_i} \mathbf{Adj}_k\right) \arrow[dr, dashed] & \\ c_k \bigsqcup_{c_i} c_k \arrow[u, "\simeq_i"] \arrow[r, "L \bigsqcup_{c_i} L"] \arrow[drr] & \mathbf{Adj}_k \bigsqcup_{c_i} \mathbf{Adj}_k \arrow[u, "\simeq_i"'] \arrow[dr, dashed] & \mathbb{L}^{\Delta}_{S,i}(X) \\ & & X \arrow[u, "\simeq_i"'] \end{tikzcd}$$

We want to show the top pentagon commutes. Because every other polytope commutes in the diagram, we have the desired lift.

- **Case $i = k$:** Fix $c_k = \Delta_{1,\dots,1,0,\dots,0} \to \mathbb{L}^{\Delta}_{S,k+1} \circ \mathbb{L}^{\Delta}_{S,k}(X)$. Because $\mathbb{L}^{\Delta}_{S,k+1}$ doesn't affect the space of k-morphisms, we can factor this map through $c_k = \Delta_{1,\dots,1,0,\dots,0} \to \mathbb{L}^{\Delta}_{S,k}(X)$. We then reproduce the same argument, keeping the first diagrm identical, and replacing $\mathbb{L}^{\Delta}_{S,i}$ by $\mathbb{L}^{\Delta}_{S,k+1} \circ \mathbb{L}^{\Delta}_{S,k}$ everywhere in the second diagram.

■

Sometimes, in situations such as Proposition 2.15, one needs to show that the Segalification of a reduced n-fold simplicial space X has adjoints even though X itself does not strictly-speaking have adjoints according to Definition 2.2. Indeed, if any k-morphism of X can be witnessed as a composition (or "pasting") of k-morphisms which do have adjoints, then the Segalification of X will have adjoints for all k-morphisms. For technical reasons, proving this for a general k is impractical as the notion of "composition" can become quite complicated, and the notion of n-fold simplicial space doesn't quite capture the stability of this notion: a composition of compositions is not easily seen as a composition itself. For the sake of this paper, it suffices to treat $k = 1$, which is significantly more elementary.

Lemma 2.6 (Partial Adjoints in Segalification)

Let X be a reduced n-fold simplicial space. Let $\mathcal{I} \subset \pi_0 X_{1,0,\dots,0}$ be the minimal subset such that:

- *if $f\colon c_1 \to X$ has a right (resp. left) adjoint, as defined in Definition 2.2, f in $\pi_0 X_{1,0,\dots,0}$ lies in $\mathcal{I}$,*
- *if $f\colon c_1 \to X$ can be written as a composition of morphisms in $\mathcal{I}$, then f lies in $\mathcal{I}$: in other words, if we have $\Delta_{r,0\dots,0} \twoheadrightarrow c_1$ preserving end-points, and, if for all consecutive inclusions $c_1 \hookrightarrow \Delta_{r,0\dots,0}$, the composition lies in $\mathcal{I}$, then f lies in $\mathcal{I}$.*

$$\begin{tikzcd}
c_1 \arrow[d, "\forall"'] \arrow[dr, "\in\mathcal{I}"] & \\
\Delta_{r,0\dots,0} \arrow[r] \arrow[d] & X \\
c_1 \arrow[ur, "f"'] &
\end{tikzcd}$$

Suppose $\mathcal{I} = \pi_0 X_{1,0,\dots,0}$ is the whole set. Then $\mathbb{L}^{\Delta}_{S}(X)$ has right (resp. left) adjoints for all 1-cells.

Proof: The proof is almost identical to second part of the proof above. One can simply observe with the same diagram that any 1-cell in $\mathcal{I}$ has an adjoint in the Segalification. ■

Recall that our goal here is to show that a colimit of (∞, n)-categories with adjoints also has adjoints. As explained in the beginning of the section, this require both to show a compatibility with Segalification (as done in the results above) and with completion, which is done bellow.

Lemma 2.7 (Completion Preserves Adjoints)

Let X be an n-fold Segal space. Let $c_k \to \mathbb{L}_C X$ be a k-cell in the completion of X. There exists a k-cell of $c_k \to X$ factoring $c_k \to \mathbb{L}_C X$.

Proof: Note that by (2.1), one prove this result for Θ^n-spaces instead of n-fold Segal spaces. By [AF18, Prop. 2.4], we have an explicit formula for the univalent completion:

$$[c_k, \mathbb{L}_C X] = \varinjlim_{\Theta_n} [c_k \times E(-), X]$$

where $E\colon \Theta_n \to \mathcal{P}\mathrm{sh}(\Theta_n)$ sends the $T \in \Theta_n$ to the free strict n-groupoid generated by T (see [AF18, Def. 2.1] for more details). In particular, for any $c_k \to \mathbb{L}_C X$, we have a factorization

$$\begin{tikzcd}
 & c_k \arrow[r] & \mathbb{L}_C X \\
c_k \times E(\bullet) \arrow[ur, equal] \arrow[r, "\simeq_C"'] & c_k \times E(T) \arrow[u, "\simeq_C"'] \arrow[r] & X \arrow[u, "\simeq_C"']
\end{tikzcd}$$

■

We can now finally prove the main result of this subsection:

Proposition 2.8 (Colimits have Adjoints)

Let $F\colon I \to \mathbf{Cat}_{(\infty,n)}$ be a diagram of (∞, n)-categories which all admit right (resp. left) adjoints for all k-morphisms. Then so does $\varinjlim F$.

Proof: By (2.1), one can compute the colimit as $\mathbb{L}_C \circ \mathbb{L}_S^{\Delta}\left(\varinjlim F\right)$ where the colimit is computed in $\mathcal{P}\mathrm{sh}(\Delta^n)$. By definition, the collection of simplicial spaces with right (resp. left) adjoints for all k-morphisms is closed under colimits in $\mathcal{P}\mathrm{sh}(\Delta^n)$. By Lemma 2.5, $\mathbb{L}_S^{\Delta}\left(\varinjlim F\right)$ has right (resp. left) adjoints for all k-morphisms. Fix $c_k \to \mathbb{L}_C \circ \mathbb{L}_S^{\Delta}\left(\varinjlim F\right)$ a k-morphism. By Lemma 2.7, we have a factorization through $c_k \to \mathbb{L}_S^{\Delta}\left(\varinjlim F\right)$ which then has an adjoint. ■

2.2 Freely Adding Adjoints and Subcategories with Adjoints

Using the results of the previous subsection, we now establish some basic "folklore" results about the (∞, n)-categories with adjoints, which we define bellow. The most important result of this section is Proposition 2.11.

Definition 2.3 (Categories with Adjoints)**.** Let $S \subset \{1, \ldots, n\}$ and $\mathcal{A}$: $\mathbf{Cat}_{(\infty,n+1)}$. We say that $\mathcal{A}$ has S-adjoints if, for all $k \in S$ and all $c_k \to \mathcal{A}$, we have lifts

$$\begin{tikzcd}
\mathbf{Adj}_k \arrow[dr, dashed] & \\
c_k \arrow[u, "L"] \arrow[r] \arrow[d, "R"'] & \mathcal{A} \\
\mathbf{Adj}_k \arrow[ur, dashed] &
\end{tikzcd}$$

Denote $\mathbf{Cat}^{S\text{-adj}}_{(\infty,n+1)} \hookrightarrow \mathbf{Cat}_{(\infty,n+1)}$ the full subcategory of $(\infty, n+1)$-categories with S-adjoints.

Note that by the Corollary 2.2, these lifts are unique if they exists. Before we head into the proof of Proposition 2.11, we observe a couple small results. Note that by Corollary 2.2, we find:

Corollary 2.9

Let $k \geq 0$. One can write $\mathbf{Adj}_k$ as a finite colimit diagram of the c_i's.

Proof: Existence of L, R, η, ϵ and the commutativity of the triangle identities can easily be encoded by some finite diagram. The Theorem above shows that the canonical cocone on $\mathbf{Adj}_k$ that one gets is colimiting. ■

Lemma 2.10 (Stability under Limits)

Let $S \subset K \subset \{1, \ldots, n\}$. The full subcategory $\mathbf{Cat}^{S\text{-adj}}_{(\infty,n+1)} \hookrightarrow \mathbf{Cat}_{(\infty,n+1)}$ is closed under all small limits (that exists) in $\mathbf{Cat}_{(\infty,n+1)}$.

Proof: Let $F\colon I \to \mathbf{Cat}^{S\text{-adj}}_{(\infty,n+1)}$ be a diagram. For $k \in S$ and assume the limit of F exists in $\mathbf{Cat}_{(\infty,n+1)}$. Then by Corollary 2.2, we have a commutative diagram

$$\begin{tikzcd}
{[\mathbf{Adj}_k, \varprojlim_{i:\,\mathcal{I}} F(i)]} \arrow[r, hook] \arrow[d, "\simeq"'] & {[c_k, \varprojlim_{i:\,\mathcal{I}} F(i)]} \arrow[d, "\simeq"] \\
\varprojlim_{i:\,\mathcal{I}} [\mathbf{Adj}_k, F(i)] \arrow[r, "\simeq"'] & \varprojlim_{i:\,\mathcal{I}} [c_k, F(i)]
\end{tikzcd}$$

where the bottom horizontal map is an equivalence, which gives that the top horizontal map is an equivalence. This means that $\varprojlim F$ has S-adjoints. ■

Using the two results above, and our main result of the previous subsection (Proposition 2.8), we can then show the existence of the full-subcategories with adjoints.

Proposition 2.11 (Free and Sub-Categories with Adjoints)

Let $n \geq 0$ and $S \subset \{1, \ldots, n\}$. The inclusion $\mathbf{Cat}^{S\text{-adj}}_{(\infty,n+1)} \hookrightarrow \mathbf{Cat}_{(\infty,n+1)}$ fits into a bireflection between locally finitely-presentable categories:

$$\begin{tikzcd}
\mathbf{Cat}^{S\text{-adj}}_{(\infty,n+1)} \arrow[r, hook, "\perp"] & \mathbf{Cat}_{(\infty,n+1)} \arrow[l, bend right, "\mathsf{Free}^S"'] \arrow[l, bend left, "\mathsf{Sub}^S"] \arrow[l, phantom, "\perp"]
\end{tikzcd}$$

Furthermore, the unit of the top adjunction $\mathcal{C} \twoheadrightarrow \mathsf{Free}^S(\mathcal{C})$ is an epimorphism and the counit of the bottom adjunction $\mathsf{Sub}^S(\mathcal{C}) \hookrightarrow \mathcal{C}$ is a monomorphism.

Proof: $\mathbf{Cat}_{(\infty,n+1)}$ has small colimits and is locally small. Note that $\mathbf{Cat}^{S\text{-adj}}_{(\infty,n+1)}$ is the category of $\mathcal{W}$-weakly local objects [Lur18, Def. 04LH] for the small collection $\mathcal{W} = \{c_k \overset{L}{\twoheadrightarrow} \mathbf{Adj}_k \mid k \in S\} \cup \{c_k \overset{R}{\twoheadrightarrow} \mathbf{Adj}_k \mid k \in S\}$. The domain of these maps of $\mathcal{W}$ are all c_k and hence compact. The codomains of these maps are all $\mathbf{Adj}_k$'s and hence compact by Corollary 2.9. By the Small Object Argument [Lur18, Cor. 04MD], the inclusion $\mathbf{Cat}^{S\text{-adj}}_{(\infty,n+1)} \hookrightarrow \mathbf{Cat}_{(\infty,n+1)}$ has a left adjoint Free^S.

Note that this implies $\mathbf{Cat}^{S\text{-adj}}_{(\infty,n+1)}$ has all colimits and that they are preserved by Free^S. Explicitly, these colimits are computed by taking the diagram in $\mathbf{Cat}_{(\infty,n+1)}$ via the inclusion, taking its colimit there, and then applying Free^S. By Proposition 2.8, the last step is not necessary: the colimit taken in $\mathbf{Cat}_{(\infty,n+1)}$ already has S-adjoints. Hence, the inclusion preserves all small colimits and thus is accessible. This shows that $\mathbf{Cat}^{S\text{-adj}}_{(\infty,n+1)}$ is a locally finitely-presentable category. By the Adjoint Functor Theorem [Lur18, Thm. 06Q5], the inclusion $\mathbf{Cat}^{S\text{-adj}}_{(\infty,n+1)} \hookrightarrow \mathbf{Cat}_{(\infty,n+1)}$ has a right adjoint Sub^S.

Because epimorphisms are stable under pushout and transfinite composition, the unit $\mathcal{C} \twoheadrightarrow \mathsf{Free}^S(\mathcal{C})$ is an epimorphism by construction. For $\mathcal{A}, \mathcal{B} \in \mathbf{Cat}_{(\infty,n+1)}$, we then have $[\mathcal{A}, \mathsf{Sub}^S(\mathcal{B})] \simeq [\mathsf{Free}^S(\mathcal{A}), \mathsf{Sub}^S(\mathcal{B})] \simeq [\mathsf{Free}^S(\mathcal{A}), \mathcal{B}] \hookrightarrow [\mathcal{A}, \mathcal{B}]$, showing that the counit $\mathsf{Sub}^S(\mathcal{B}) \hookrightarrow \mathcal{B}$ is a monomorphism. ■

One can easily build upon this and find a *relative* version of the result above.

Corollary 2.12

Let $S \subset K \subset \{1, \dots, n\}$. The inclusion $\mathbf{Cat}^{K\text{-adj}}_{(\infty,n+1)} \hookrightarrow \mathbf{Cat}^{S\text{-adj}}_{(\infty,n+1)}$ fits into a bireflection between locally finitely-presentable categories:

$$\begin{tikzcd} \mathbf{Cat}^{K\text{-adj}}_{(\infty,n+1)} \arrow[r, hook] \arrow[r, phantom, shift left=3, "\perp"] \arrow[r, phantom, shift right=3, "\perp"] & \mathbf{Cat}^{S\text{-adj}}_{(\infty,n+1)} \arrow[l, bend right, "\mathsf{Free}^{S\to K}"'] \arrow[l, bend left, "\mathsf{Sub}^{S\to K}"] \end{tikzcd}$$

Furthermore, the unit of the top adjunction $\mathcal{C} \twoheadrightarrow \mathsf{Free}^{S\to K}(\mathcal{C})$ is an epimorphism and the counit of the bottom adjunction $\mathsf{Sub}^{S\to K}(\mathcal{C}) \hookrightarrow \mathcal{C}$ is a monomorphism.

Proof: The argument is entirely similar to the one of Proposition 2.11. We can identify $\mathbf{Cat}^{K\text{-adj}}_{(\infty,n+1)} \hookrightarrow \mathbf{Cat}^{S\text{-adj}}_{(\infty,n+1)}$ with the full subcategory of $\mathcal{W}$-weakly local objects for the small collection $\mathcal{W} = \{\mathsf{Free}^S(c_k \overset{L}{\twoheadrightarrow} \mathbf{Adj}_k) \mid k \in K\} \cup \{\mathsf{Free}^S(c_k \overset{R}{\twoheadrightarrow} \mathbf{Adj}_k) \mid k \in K\}$. The functor Free^S preserves compact objects (its right adjoint preserves filtered colimit) and epimorphisms (as it preserves pushouts), so we can apply the same argument as above. ■

We now focus on a couple results which are curcial to the inductive formulation of the main theorems of this paper.

Lemma 2.13 (Free preserves (some) Ajoints)

Let $S \subset \{2, \dots, n\}$ and $m < \min(S)$. Then $\mathsf{Free}^{\emptyset\to S}$ sends $\mathbf{Cat}^{m\text{-adj}}_{(\infty,n+1)}$ to $\mathbf{Cat}^{S\cup\{m\}\text{-adj}}_{(\infty,n+1)}$.

Proof: Let $\mathcal{C} \in \mathbf{Cat}^{m\text{-adj}}_{(\infty,n+1)}$. Recall reflection $\mathcal{C} \twoheadrightarrow \mathsf{Free}^{\emptyset\to S}(\mathcal{C})$ is a transfinite composition of κ-many pushouts of maps in $\{c_k \overset{L}{\to} \mathbf{Adj}_k \mid k \in S\} \cup \{c_k \overset{R}{\to} \mathbf{Adj}_k \mid k \in S\}$, for some ordinal κ. We show by transfinite induction that the codomain of any such composition lands in $\mathbf{Cat}^{m\text{-adj}}_{(\infty,n+1)}$.

- **Initialization:** By definition, $\mathcal{C}_0 = \mathcal{C} \colon \mathbf{Cat}^{m\text{-adj}}_{(\infty,n+1)}$.
- **Successor step:** Let $\alpha < \kappa$. There is a $k \in S$ and a pushout diagram

$$\begin{tikzcd} c_k \arrow[r] \arrow[d] & \mathbf{Adj}_k \arrow[d, dashed] \\ \mathcal{C}_\alpha \arrow[r, dashed] \arrow[ur, phantom, "\ulcorner"] & \mathcal{C}_{\alpha+1} \end{tikzcd}$$

By Lemma 2.7, it suffices to show the m-morphisms $c_m \to \mathbb{L}^{\Delta}_S(\mathcal{C}_\alpha \sqcup_{c_k} \mathbf{Adj}_k)$ have adjoints. It is immediate that for $a_1, \dots, a_m \geq 0$, we have $(c_k)_{a_1,\dots,a_m,0,\dots,0} \simeq (\mathbf{Adj}_k)_{a_1,\dots,a_m,0,\dots,0}$. Hence, we have an equivalence $(\mathcal{C}_\alpha)_{a_1,\dots,a_m,0,\dots,0} \simeq (\mathcal{C}_\alpha \sqcup_{c_k} \mathbf{Adj}_k)_{a_1,\dots,a_m,0,\dots,0}$. So the pushout is already Segal in the first m-directions and in the iterative Necklace formula [BS24, Thm. D], the first m steps do nothing. The last steps do not affect the space of m-morphisms either, so we have $(\mathcal{C}_\alpha)_{a_1,\dots,a_m,0,\dots,0} \simeq (\mathbb{L}^{\Delta}_S(\mathcal{C}_\alpha \sqcup_{c_k} \mathbf{Adj}_k))_{a_1,\dots,a_m,0,\dots,0}$. By assumption, every m-morphism in $\mathcal{C}_\alpha$ has an adjoint, which concludes the successor step.

- **Limit step:** Let $\lambda \leq \kappa$ be a limit ordinal. We can write $\mathcal{C}_\lambda = \varinjlim_{\alpha<\lambda} \mathcal{C}_\alpha$, a colimit of categories with m-adjoints. By Lemma 2.10, we have $\mathcal{C}_\lambda \colon \mathbf{Cat}^{m\text{-adj}}_{(\infty,n+1)}$.

By the transfinite induction principle, we get $\mathsf{Free}^{\emptyset\to S}(\mathcal{C}) \colon \mathbf{Cat}^{m\text{-adj}}_{(\infty,n+1)} \cap \mathbf{Cat}^{S\text{-adj}}_{(\infty,n+1)} = \mathbf{Cat}^{S\cup\{m\}\text{-adj}}_{(\infty,n+1)}$. ■

Corollary 2.14 (Composing Free/Sub Constructions)

Let $k < n$ and $S \subset \{k+1, \dots, n\}$. Then $\mathsf{Free}^{\{k\}\cup S} = \mathsf{Free}^S \circ \mathsf{Free}^{\{k\}}$.

Proof: Let $\mathcal{A} \in \mathbf{Cat}_{(\infty,n+1)}$. By Lemma 2.13, we know that $\mathsf{Free}^S \circ \mathsf{Free}^{\{k\}}(\mathcal{A})$ has $\{k\} \cup S$-adjoints. The prove the first clim, it suffices to see that it satifies the correct universal property. ■

2.3 Lifting Once-Fully-Adjoint Morphisms

In this subsection, we finalize the categorical prerequisites for the inductive argument at the core of the proof of the main theorem of this paper (Theorem 4.2). The result bellow formalizes the idea that "constucting a full-$\{1\}$-adjoints is equivalent to finding a sequence of adjunctions"

Proposition 2.15 (Free Full Adjunction)

Denote $\mathbb{Z}^{\to\leftarrow}$ the poset whose objects are the integers and which has morphisms $2n \to 2n \pm 1$ for all $n \colon \mathbb{Z}$. We have a colimit diagram

$$\begin{tikzcd}
 & & & \mathsf{Free}^{\{k\}}(c_k) & & & \\
\cdots \arrow[rrrr, dashed] & & \mathbf{Adj}_k \arrow[rrr, dashed] & & \mathbf{Adj}_k \arrow[u, dashed] & & \mathbf{Adj}_k \arrow[lll, dashed] & & \cdots \arrow[llll, dashed] \\
 & c_k \arrow[ul, "R"] \arrow[ur, "L"'] & & c_k \arrow[ul, "R"] \arrow[ur, "L"'] & & c_k \arrow[ul, "R"] \arrow[ur, "L"'] & & c_k \arrow[ul, "R"] \arrow[ur, "L"'] &
\end{tikzcd}$$

We denote by $f^{\vee n} \colon c_k \to \mathsf{Free}^{\{k\}}(c_k)$ the n-th cocone projection (from left to right, mneaning that $\vee$ correspond to taking right adjoints), which are all epimorphisms.

Proof: It suffices to show the result for $k = 1$, as by Remark 2.2, suspension is the composition of a left adjoint and the forgetful from the slice, and hence commutes with contractible colimits. Firstly, we build the cocone diagram. By definition, we know that $\mathsf{Free}^{\{1\}-\mathsf{adj}}(c_1)$ is fully-$\{1\}$-adjoint and receives a universal functor $c_1 \to \mathsf{Free}^{\{1\}-\mathsf{adj}}(c_1)$. The space of cocones is the limit of the diagram $\mathbb{Z}^{\to\leftarrow} \to \mathcal{S}$ below:

$$\begin{tikzcd}
\cdots \arrow[dr, "R^*"] & & [\mathbf{Adj}, \mathsf{Free}^{\{1\}-\mathsf{adj}}(c_1)] \arrow[dl, "L^*"'] \arrow[dr, "R^*"] & & \cdots \arrow[dl, "L^*"'] \\
 & [c_1, \mathsf{Free}^{\{1\}-\mathsf{adj}}(c_1)] & & [c_1, \mathsf{Free}^{\{1\}-\mathsf{adj}}(c_1)] &
\end{tikzcd}$$

By Theorem 2.1, this is an infinite zig-zag of weak homotopy equivalences. By picking the point in the zero-th space corresponding to the universal functor $c_1 \to \mathsf{Free}^{\{1\}-\mathsf{adj}}(c_1)$, we get a canonical cocone diagram.

Secondly, we build a functor $\mathsf{Free}^{\{1\}-\mathsf{adj}}(c_1) \to \varinjlim(\mathbb{Z}^{\to\leftarrow} \to \mathbf{Cat}_{(\infty,2)})$. By universal property of $\mathsf{Free}^{\{1\}-\mathsf{adj}}(c_1)$, it is sufficient to show that $\varinjlim(\mathbb{Z}^{\to\leftarrow} \to \mathbf{Cat}_{(\infty,2)})$ has $\{1\}$-adjoints and to pick a 1-morphism $c_1 \to \varinjlim(\mathbb{Z}^{\to\leftarrow} \to \mathbf{Cat}_{(\infty,2)})$. The first part is a consequence of Lemma 2.6 and Lemma 2.7, as any morphism coming from a copy of $\mathbf{Adj}$ is given by composition of L's and R's, which have adjoints. The second part follows by choosing the 1-cell $c_1 \to \varinjlim(\mathbb{Z}^{\to\leftarrow} \to \mathbf{Cat}_{(\infty,2)})$ given by the universal cocone projection on the zero-th copy of c_1.

It is now easy to see that the two functors $\mathsf{Free}^{\{1\}-\mathsf{adj}}(c_1) \to \varinjlim(\mathbb{Z}^{\to\leftarrow} \to \mathbf{Cat}_{(\infty,2)})$ and $\varinjlim(\mathbb{Z}^{\to\leftarrow} \to \mathbf{Cat}_{(\infty,2)}) \to \mathsf{Free}^{\{1\}-\mathsf{adj}}(c_1)$ are inverses to each other by examining the universal properties. ■

Remark 2.3. This result can be generalized to construct any $\mathsf{Free}^{\{k,\dots,n\}-\mathsf{adj}}(c_k)$ as a colimit. This however requires a more careful generalization of Proposition 2.8. One can even proof that for $k < n$, the colimit diagrams can be chosen finite, by carefully looking at the notion of redundancy of adjunction data (see Lemma 4.1 for an example of such a result).

We get the following corollary, which will be used in the proof of the main theorem of this paper (Theorem 4.2).

Corollary 2.16 (Lifting Once-Fully-Dualizable Morphisms)

Let $1 \leq k < n$. Suppose we have a monomoprhism $\mathcal{A} \hookrightarrow \mathcal{B} \colon \mathbf{Cat}_{(\infty,n)}$, a map $\mathsf{Free}^{\{k\}}(c_k) \to \mathcal{B}$ such that all precomposition with $f^{\vee n} \colon c_k \to \mathsf{Free}^{\{k\}}(c_k)$ lifts to $\mathcal{A}$ as in the diagram below. Then, there exists a dashed lift if and only if, for all n, there exists the dotted lifts:

$$\begin{tikzcd}
 & \mathcal{A} \arrow[rr, hook] & & \mathcal{B} \\
c_{k+1} \arrow[ur, dotted] \arrow[dr] & & & \\
 & c_k \arrow[uu, hook] \arrow[r, two heads] \arrow[uur, hook, dashed] & \mathsf{Free}^{\{k\}}(c_k) \arrow[uu] & \\
c_{k+1} \arrow[uur, dotted] \arrow[ur] & & &
\end{tikzcd}$$

$\epsilon(f^{\vee n})$, $f^{\vee n}$, $\eta(f^{\vee n})$

In other words: the fully-$\{k\}$-dualizable morphism lift if all of its units and counits lift.

2.4 Invertibility

In this final subsection, we compute the clssifying spaces of free/walking adjunctions. Contractibility of these spaces exactly correspond to showing that any invertible morphisms is fully-adjointable, which is part of the statement of Theorem 4.5.

Lemma 2.17 (Classifying Space of Adj)

The localization of **Adj** *by freely inverting η and ϵ is the contractible category.*

Proof: In $\mathbf{Cat}_{(\infty,n)}$, the terminal functor $c_1 \to \{\bullet\}$ can be seen as a localization or truncation. Indeed, using completeness, precompositing by this map induces the inclusion $[c_1, \mathcal{C}]_{\simeq} \hookrightarrow [c_1, \mathcal{C}]$ of the connected component of all invertible 1-morphisms (for example, see [Rez01, Thm. 6.2 and Prop. 6.4]). We have a factorization $c_1 \twoheadrightarrow \mathbf{Adj} \twoheadrightarrow \mathbf{Adj}[\eta^{-1}, \epsilon^{-1}] \to \{\bullet\}$ where the composition is epimorphic. This implies that $\mathbf{Adj}[\eta^{-1}, \epsilon^{-1}] \twoheadrightarrow \{\bullet\}$ is epimorphic as well and hence, by applying any $[-, \mathcal{C}]$, we get

$$[c_1, \mathcal{C}]_{\simeq} \hookrightarrow [\mathbf{Adj}[\eta^{-1}, \epsilon^{-1}], \mathcal{C}] \hookrightarrow [\mathbf{Adj}, \mathcal{C}] \hookrightarrow [c_1, \mathcal{C}]$$

To show that $\mathbf{Adj}[\eta^{-1}, \epsilon^{-1}] \twoheadrightarrow \{\bullet\}$ is an equivalence, it is sufficient to show that: given any adjunction $L \colon a \underset{\longleftarrow}{\overset{\longrightarrow}{\perp}} b : R \colon \mathcal{C}$, if η and ϵ are invertible, then L is invertible. Invertibility is a property detected by truncation to the homotopy 1-category, where this is a classical result of ordinary category theory. ■

Similarly, we get the following:

Lemma 2.18 (Classifying Space of Higher Walking Adjoints)

Let $1 \leq k < n$ and $S \subset \{k, \ldots, n-1\}$. The terminal functor $\mathsf{Free}^S(c_k) \twoheadrightarrow c_{k-1}$ is an epimorphism and exhibits c_{k-1} as the localization at all i-morphisms for $i > k$, meaning that it is the reflection map of the localization $|-|_{>k-1} \colon \mathbf{Cat}_{(\infty,n)} \to \mathbf{Cat}_{(\infty,k-1)}$.

Proof: As above, the composition $c_k \twoheadrightarrow \mathsf{Free}^S(c_k) \to \{\bullet\}$ is epimorphic, implying the first part of the statement. We proceed by downward induction on $|S$.

- **Base case $|S| = 1$:** If $k \notin S$ then $\mathsf{Free}^S(c_k) = c_k \twoheadrightarrow c_{k-1}$ and the result is a consequence of completeness. Otherwise, the truncation functor $|-|_{>k-1}$ being left adjoint will preserve the colimit in Proposition 2.15. The diagram being sent to the terminal $n \in \mathbb{Z}^{\to\leftarrow} \mapsto \{\bullet\}$ diagram, we get $|(\mathsf{Free}^{\{k\}}(c_k)|_{>k-1} \simeq \{\bullet\}$, as $\mathbb{Z}^{\to\leftarrow}$ is contractible (this can be seen by Quillen's Theorem A).
- **Inductive step:** Let $|S| > 1$. As above, one can assume without loss of generality that $S = \{k\} \sqcup S'$. Similarly to the argument of Lemma 2.17, we want to show that the inclusion $[c_k, \mathcal{C}]_{\simeq} \hookrightarrow [|\,\mathsf{Free}^S(c_k)|_{>k-1}, \mathcal{C}]$ is an equivalence. By Corollary 2.14, this is equivalent to showing $[c_k, \mathcal{C}]_{\simeq} \hookrightarrow [\mathsf{Free}^{\{k\}}(c_k), \mathsf{Sub}^{S'}(\mathcal{C})]$ ■

3 Dualizability Data as Factorization Homology

The goal of this section is to establish the algebraic and geometric elements of the proof of our criteria. In subsection 3.1 and subsection 3.2, we use factorization homology [AFT17b, AFR18b, AFT17a] to construct the iterated "(co)units of (co)units of (co)units of ...", which we call adjunction data. In subsection 3.3, we prove

a criterion for existence of adjoints of top-level morphisms in the higher Morita categories (see [Lur09a, Sch14, Hau17b, Hau23] for its definition). Finally in subsection 3.4, we identify some units in the dualizability data with universal algebra-map to some higher Hochschild Cohomology.

Throughout the rest of this paper, we fix $\mathcal{V} \in \mathbf{Alg}_{\mathcal{E}_\infty}\left(\mathbf{Cat}_{(\infty,1)}^{\mathsf{geom}-\xrightarrow{\text{colim}}}\right)$, a symmetric monoidal $(\infty,1)$-category with geometric realizations and whose monoidal structure distributes over geometric realizations.

3.1 Homological Commutative Diagrams

This first subsection establishes a key technical tool to *transfer* algebraic structures "carried" by different stratified spaces, which is stated in the form of Lemma 3.5. We start with quick overview of the formalism of factorization homology of conically smooth stratified spaces developed by Ayala, Francis, Rozenblyum and Tanaka in [AFT17a, AFT17b]. All the stratified spaces we will consider will be conically smooth [AFT17b, Subsec. 4.1] and finitary [AFT17b, Def. 8.3.6], so we decide to omit these adjectives in the rest of this paper. We slightly modify the original notations of [AFT17b] as follows:

- Disk will denote the symmetric monoidal $(1,1)$-category of "disks" with open embeddings: finite disjoint unions of basic stratified spaces, see [AFT17b, Subsec. 3.2.] for their inductive definition,
- Mfld will denote the symmetric monoidal $(1,1)$-category of stratified spaces with open embeddings,
- $\mathcal{D}\mathsf{isk}$ will denote the symmetric monoidal $(\infty,1)$-category obtained of disks with open embedding spaces see [AFT17b, Subsec. 4.1] (denoted $\mathcal{S}\mathsf{ngl}\mathsf{r}$ in the original text),
- $\mathcal{M}\mathsf{fld}$ will denote the symmetric monoidal $(\infty,1)$-category obtained of stratified spaces with open embedding spaces (denoted $\mathcal{S}\mathsf{ngl}\mathsf{r}$ in the original text).

We now briefly recall the operadic notions required to talk about factorization homology of stratified spaces. Once again, we slightly modify the original notations of [AFT17a] to fit our exposition.

Definition 3.1 (Marked Operad)**.** An operad $\underline{\mathcal{O}} \to \mathbf{FinSet}_*$ is said to be *marked* if it is endowed with a collection of coCartesian morphisms called *pre-coCartesian* which contains all inert morphisms. For two marked operads $\underline{\mathcal{A}}, \underline{\mathcal{B}} \to \mathbf{FinSet}_*$, the category of algebras $\mathbf{Alg}_{\underline{\mathcal{A}}/\underline{\mathcal{B}}} \hookrightarrow [\underline{\mathcal{A}}, \underline{\mathcal{B}}]_{/\mathbf{FinSet}^*}$ is the full-$(infty,1)$-subcategory of functors above $\mathbf{FinSet}^*$ on these functors which sends pre-coCartesian morphisms to pre-coCartesian morphisms.

Note that usual operads can be seen as marked operads by declaring only inert morphisms to be pre-coCartesian, and symmetric monoidal $(\infty,1)$-categories can be seen as marked operads by declaring all co-Cartesian morphisms to be pre-coCartesian. Recall from [Lur17, Subsubsection 2.4.3] that given a category $\mathcal{O}$, we can form an operad $\mathcal{O}^{\sqcup} \to \mathbf{FinSet}_*$ whose objects over $\mathbf{n}^+$ are n-uples of objects of $\mathcal{O}$. We can then directly adapt the following result from [AFT17a].

Lemma 3.1 (Pullback of Marked Operads along Right Fibrations, [AFT17a, Cor. 1.20])

Let $\underline{\mathcal{O}} \to \mathbf{FinSet}_*$ be a marked operad and $\mathcal{E} \to \mathcal{O}$ be a right fibration. The pullback

$$\begin{tikzcd}
\underline{\mathcal{E}} \arrow[r, dashed] \arrow[d, dashed] \arrow[dr, phantom, "\lrcorner", very near start] & \mathcal{E}^{\sqcup} \arrow[d] \\
\underline{\mathcal{O}} \arrow[r] \arrow[dr] & \mathcal{O}^{\sqcup} \arrow[d] \\
& \mathbf{FinSet}^*
\end{tikzcd}$$

exhibit $\underline{\mathcal{E}}$ as a marked operad with pre-coCartesian morphisms those mapping to pre-coCartesian morphisms in $\underline{\mathcal{O}}$.

Definition 3.2 ([AFT17a, Const. 2.4])**.** Using the lemma above, for X a stratified space, we define

- $\underline{\mathsf{Disk}}_{/X}$ as the marked $(1,1)$-operad obtained from the slice right fibration $\underline{\mathsf{Disk}_{/X}} \to \mathsf{Disk}$ and the symmetric monoidal structure on Disk,
- $\underline{\mathsf{Mfld}}_{/X}$ as the marked $(1,1)$-operad obtained from the slice right fibration $\underline{\mathsf{Mfld}_{/X}} \to \mathsf{Mfld}$ and the symmetric monoidal structure on Mfld,
- $\underline{\mathsf{Disk}}_{/X}$ as the marked $(\infty,1)$-operad obtained from the slice right fibration $\underline{\mathsf{Disk}_{/X}} \to \mathsf{Disk}$ and the

symmetric monoidal structure on $\mathcal{D}\mathsf{isk}$,

- $\underline{\mathsf{Mfld}}_{/X}$ as the marked $(\infty,1)$-operad obtained from the slice right fibration $\underline{\mathsf{Mfld}_{/X}} \to \mathsf{Mfld}$ and the symmetric monoidal structure on $\mathcal{M}\mathsf{fld}$,

To simplify notations, we let $\mathbf{Alg}_{/X}(\mathcal{V})$ denote $\mathbf{Alg}_{\underline{\mathcal{D}\mathsf{isk}}_{/X}/\underline{\mathcal{O}}}$ for $\underline{\mathcal{O}}$ a marked operad. More explicitly, the operad $\mathcal{D}\mathsf{isk}_{/X}$ is given as follows:

- its objects lying over $\mathbf{n}^+$ are collections $(\bigsqcup_{1\leq i\leq k_j} B_{i,j} \hookrightarrow X)_{1\leq j\leq n}$ of embeddings of a finite disjoint union of basics into X,
- the space of morphisms from $a = (\bigsqcup_{1\leq i\leq k_j} B_{i,j} \hookrightarrow X)_{1\leq j\leq n}$ to $b = (\bigsqcup_{1\leq i\leq l_j} B'_{i,j} \hookrightarrow X)_{1\leq j\leq m}$ over $\phi\colon \mathbf{n}^+ \to \mathbf{m}^+ \colon \mathbf{FinSet}_*$ of the pullback in spaces

$$\begin{tikzcd}
\mathcal{D}\mathsf{isk}(\mathcal{B})_{/X}(a,b) \arrow[rr, dashed] \arrow[d, dashed] \arrow[dr, phantom, "\rightsquigarrow"] & & \{\bullet\} \arrow[d] \\
\prod_{1\leq j'\leq m} \mathsf{Emb}(\bigsqcup_{j\in\phi^{-1}(j')} \bigsqcup_{1\leq i\leq k_j} B_{i,j}, \bigsqcup_{1\leq i\leq l_{j'}} B'_{i,j'}) \arrow[r] & \prod_{j\in\phi^{-1}(\mathbf{m})} \mathsf{Emb}(\bigsqcup_{1\leq i\leq k_j} B_{i,j}, X) &
\end{tikzcd}$$

- a morphism above is pre-coCartesian if each of the maps $\bigsqcup_{j\in\phi^{-1}(j')} \bigsqcup_{1\leq i\leq k_j} B_{i,j} \hookrightarrow \bigsqcup_{1\leq i\leq l_{j'}} B'_{i,j'})$ in the bottom left space are isotopy equivalences.

Note that these constructions are functorial in X with respect to open embeddings, as it directly follows from the slice construction which is itself functorial. In particular, we have an action $\mathbf{Aut}(X) \curvearrowright \mathcal{D}\mathsf{isk}_{/X}$ by equivalences of marked operads.

Using the equivalence of [AFT17a, Thm. 2.43], one can define homology theories as follows.

Definition 3.3 (Homology Theories [AFT17a, Def. 2.37])**.** For X a stratified space, the category of *homology theories* $\mathcal{H}(\underline{\mathcal{M}\mathsf{fld}}_{/X}, \mathcal{V})$ is the full-$(\infty,1)$-subcategory

$$\mathcal{H}(\underline{\mathcal{M}\mathsf{fld}}_{/X}, \mathcal{V}) \hookrightarrow \mathbf{Alg}_{\underline{\mathcal{M}\mathsf{fld}}_{/X}}(\mathcal{V})$$

on these morphisms of operads $\mathcal{H}$ such that the following equivalent conditions are satisfied:

- for all $Z \in \mathcal{M}\mathsf{fld}_{/X}$, the canonical morphism

$$\varinjlim\big(\mathcal{D}\mathsf{isk}_{/Z} \to \mathcal{M}\mathsf{fld}_{/X} \to \mathcal{V}\big) \xrightarrow{\sim} \mathcal{H}(Z)$$

  is an equivalence.

- for all $Z \cong Z_- \cup_{\mathbb{R}\times Z_0} Z_+$ collar-glueing over X, the canonical morphism

$$\big|\mathcal{H}(Z_-) \otimes \mathcal{H}(\mathbb{R}\times Z_0)^{\otimes\bullet} \otimes \mathcal{H}(Z_+)\big|_\bullet \xrightarrow{\sim} \mathcal{H}(Z)$$

  is an equivalence.

In the original text of [AFT17a], this definition is given for $\mathcal{V}$ a symmetric monoidal $(\infty,1)$-category with sifted colimits and whose monoidal structure distributes over sifted colimits. It's a harmless generalization to replace sifted colimits by geometric realizations, as long as we only consider *finitary* stratified spaces, meaning these obtained from basics via a sequence of collar-glueing [AFT17b, Def. 8.3.6], which is our case. Indeed, it is proven in [AFT17a, Cor. 2.38] that for a collar-glueing $X \cong X_- \cup_{\mathbb{R}\times X_0} X_+$, we have a geometric realization

$$\Big|\mathcal{D}\mathsf{isk}_{/X_-} \times \mathcal{D}\mathsf{isk}^\bullet_{/\mathbb{R}\times X_0} \times \mathcal{D}\mathsf{isk}_{/X_+}\Big|_\bullet \xrightarrow{\sim} \mathcal{D}\mathsf{isk}_{/X}$$

so that the collection of $\mathcal{D}\mathsf{isk}_{/-}$-indexed colimits is generated by geometric realizations, and thus also the collection of $\mathsf{Disk}_{/-}$-indexed colimits, as $\mathsf{Disk}_{/X} \to \mathcal{D}\mathsf{isk}_{/X}$ is a localization [AFT17a, Prop. 2.22] and hence final.

Corollary 3.2 (Disk-Algebras are the coefficients of Homology Theories)

Let X be a stratified space. By left Kan extension along the inclusion $\mathsf{Disk}_{/X} \hookrightarrow \mathcal{M}\mathsf{fld}_{/X}$, we have an

equivalence $\mathbf{Alg}_X(\mathcal{V}) \simeq \mathcal{H}(\underline{\mathcal{M}\mathsf{fld}}_{/X}, \mathcal{V})$ *fitting in the commutative diagram*

$$\begin{tikzcd}
\mathbf{Alg}_X(\mathcal{V}) \arrow[r, "\sim"] \arrow[dd] & \mathcal{H}(\underline{\mathcal{M}\mathsf{fld}}_{/X}, \mathcal{V}) \arrow[d, hook] \\
& \mathbf{Alg}_{\underline{\mathcal{M}\mathsf{fld}}_{/X}}(\mathcal{V}) \arrow[d] \\
\mathbf{Alg}_{\underline{\mathsf{Disk}}_{/X}}(\mathcal{V}) \arrow[r] & \mathbf{Alg}_{\underline{\mathsf{Mfld}}_{/X}}(\mathcal{V})
\end{tikzcd}$$

Proof Sketch: This is a direct consequence of [AFT17a, Thm. 2.43]. The details of how the left Kan extension behaves in the context of marked operads can be found in [AFT17a, Lem. 2.16 and 2.17]. ■

We can then define a convenient way to transfer between homology theories. This definition is a small abstraction from the behavior of constructible bundles in [AFT17a, Subsec. 2.5].

Definition 3.4 (Homological Morphisms)**.** Let X, Y be stratified spaces together with a marked operad morphism $\alpha\colon \underline{\mathsf{Mfld}}_{/Y} \to \underline{\mathsf{Mfld}}_{/X}$. We say that this morphism is *homological* if

(i) the functor $\mathsf{Disk}_{/Y} \xrightarrow{\mathcal{D}\mathsf{isk}_{/\alpha(-)}} \mathbf{Cat}_{(\infty,1)}$ sends isotopy equivalences to equivalences and,

(ii) the following equivalent conditions hold:

(a) for all $V \in \mathsf{Mfld}_{/Y}$, the canonical functor bellow is an equivalence

$$\underrightarrow{\mathrm{colim}}\left(\mathsf{Disk}_{/V} \xrightarrow{\mathcal{D}\mathsf{isk}_{/\alpha(-)}} \mathbf{Cat}_{(\infty,1)}\right) \xrightarrow{\sim} \mathcal{D}\mathsf{isk}_{/\alpha(V)}$$

(b) for all $V \in \mathsf{Disk}_{/Y}$, the canonical projection $X_{V,\alpha} \xrightarrow{(\star)} \mathsf{Disk}_{/\alpha(V)}$ from the *comma* $(1,1)$-category bellow is $(\infty,1)$-final

$$\begin{tikzcd}
\mathsf{Disk}_{/V} \arrow[r, "\alpha"] & \mathsf{Mfld}_{/\alpha(V)} \\
X_{V,\alpha} \arrow[u, dashed] \arrow[r, dashed, "(\star)"'] \arrow[ur, phantom, "\Nwarrow"] & \mathsf{Disk}_{/\alpha(V)} \arrow[u, hook]
\end{tikzcd}$$

In the notations of [AFT17a, Subsec. 2.5], the comma category $X_{V,\alpha}$ would be described as the limit of the diagram bellow.

$$\begin{tikzcd}
& X_{V,\alpha} \arrow[dl, dashed, "(\star)"'] \arrow[d, dashed] \arrow[dr, dashed] & \\
\mathsf{Disk}_{/\alpha(V)} \arrow[dr, hook] & [c_1, \mathsf{Mfld}_{/\alpha(V)}] \arrow[dl, "\mathsf{dom}"] \arrow[dr, "\mathsf{cod}"'] & \mathsf{Disk}_{/V} \arrow[dl] \\
& \mathsf{Mfld}_{/\alpha(V)} & \mathsf{Mfld}_{/\alpha(V)}
\end{tikzcd}$$

Proof of the Equivalence of Conditions: The proof follow from Joyal's $(\infty,1)$-categorical version of Quillen's Theorem A [Lur18, Thm. 02NY]. Details can be found in the proof of [AFT17a, Lem. 2.27], whose statement is essentially identical. ■

The proof of [AFT17a, Thm. 2.25] can be adapted in our context to show the following, by using the "universal" algebra $A = \mathcal{D}\mathsf{isk}_{/-}$ [AFT17a, Cor. 2.38].

Lemma 3.3 (Composition stays Homological)

Homological morphisms are closed under composition.

Proof: Let X, Y, Z be stratified spaces and $\underline{\mathsf{Mfld}}_{/Z} \xrightarrow{\beta} \underline{\mathsf{Mfld}}_{/Y} \xrightarrow{\alpha} \underline{\mathsf{Mfld}}_{/X}$ be composable homological morphisms. Let $W \in \mathsf{Mfld}_{/Z}$. Consider the diagram bellow.

$$\begin{tikzcd}
\mathsf{Disk}_{/W} \arrow[r, "\beta"] & \mathsf{Mfld}_{/\beta(W)} \arrow[dr, "\mathcal{D}\mathsf{isk}_{/-}"] & \\
X_{W,\beta} \arrow[u, dashed] \arrow[r, dashed, "(\star)"'] \arrow[ur, phantom, "\Leftarrow"] & \mathsf{Disk}_{/\beta(W)} \arrow[u, hook] \arrow[r, "\mathcal{D}\mathsf{isk}_{/-}"'] \arrow[ur, phantom, "\Leftarrow"] & \mathbf{Cat}_{(\infty,1)}
\end{tikzcd}$$

As in the proof of [AFT17a, Thm. 2.25], we note that the right square is a comma and the left triangle is a left Kan extension, implying that the outer triangle is also a left Kan extension. In particular, we have

$$\varinjlim\left(\mathsf{Disk}_{/W} \xrightarrow{\mathcal{D}\mathsf{isk}_{/\beta(-)}} \mathbf{Cat}_{(\infty,1)}\right) \xrightarrow{\sim} \varinjlim\left(X_{W,\beta} \to \mathsf{Disk}_{/\beta(W)} \xrightarrow{\mathcal{D}\mathsf{isk}_{/-}} \mathbf{Cat}_{(\infty,1)}\right) \xrightarrow{\sim} \varinjlim\left(\mathsf{Disk}_{/\beta(W)} \xrightarrow{\mathcal{D}\mathsf{isk}_{/-}} \mathbf{Cat}_{(\infty,1)}\right)$$

This gives the commutative diagram bellow, written with subscript notations to lighten the expressions.

$$\begin{tikzcd}
\varinjlim_{W' \subseteq W}\left(\varinjlim_{V \subseteq \beta(W')} \mathcal{D}\mathsf{isk}_{/\alpha(V)}\right) \arrow[r, "\simeq", "\alpha \text{ hom.}"'] \arrow[d, "\simeq", "\beta \text{ hom.}"'] & \varinjlim_{W' \subseteq W} \mathcal{D}\mathsf{isk}_{/\alpha\circ\beta(W')} \arrow[d] \\
\varinjlim_{V \subset \beta(W)} \mathcal{D}\mathsf{isk}_{/\alpha(V)} \arrow[r, "\simeq", "\alpha \text{ hom.}"'] & \mathcal{D}\mathsf{isk}_{/\alpha\circ\beta(W)}
\end{tikzcd}$$

where the horizontal functors are equivalences by property $(ii.a)$ for α. Thus the right vertical functor is an equivalence as well. We are left to prove that $\mathsf{Disk}_{/Z} \xrightarrow{\mathcal{D}\mathsf{isk}_{/\alpha\circ\beta(-)}} \mathbf{Cat}_{(\infty,1)}$ sends isotopy equivalences to equivalences. If $W \hookrightarrow W' \colon \mathsf{Disk}_{/Z}$ is an isotopy equivalence, then by property (i) for β the induced functor $\mathcal{D}\mathsf{isk}_{/\beta(W)} \to \mathcal{D}\mathsf{isk}_{/\beta(W')}$ is an equivalence. By property (i) for α, the colimits computed in the bottom left of the square above for W (resp. W') fit in the following diagram

$$\begin{tikzcd}
\mathsf{Disk}_{/\beta(W)} \arrow[r, "(\star)"] \arrow[d] & \mathcal{D}\mathsf{isk}_{/\beta(W)} \arrow[d, "\wr"] \arrow[dr, "\mathcal{D}\mathsf{isk}_{/\alpha(-)}"] & \\
\mathsf{Disk}_{/\beta(W')} \arrow[r, "(\star)"'] & \mathcal{D}\mathsf{isk}_{/\beta(W)} \arrow[r, "\mathcal{D}\mathsf{isk}_{/\alpha(-)}"'] & \mathbf{Cat}_{(\infty,1)}
\end{tikzcd}$$

where thhe functors marked by $(\star)$ are final and the one marked by $\sim$ is an equivalence. Thus the induced morphism of colimits is an equivalence as well, and $\alpha \circ \beta$ is indeed homological. ■

The following result justifies our naming convention for "homological" morphisms.

Lemma 3.4 (Lifting Homology Theories)

Let X, Y be stratified spaces and $\alpha \colon \underline{\mathsf{Mfld}}_{/Y} \to \underline{\mathsf{Mfld}}_{/X}$ be homological. Then, we have a unique lift

$$\begin{tikzcd}
\mathcal{H}(\underline{\mathcal{M}\mathsf{fld}}_{/X}, \mathcal{V}) \arrow[r, dashed] \arrow[d] & \mathcal{H}(\underline{\mathcal{M}\mathsf{fld}}_{/Y}, \mathcal{V}) \arrow[d] \\
\mathbf{Alg}_{\underline{\mathsf{Mfld}}_{/X}}(\mathcal{V}) \arrow[r] & \mathbf{Alg}_{\underline{\mathsf{Mfld}}_{/Y}}(\mathcal{V})
\end{tikzcd}$$

Proof: If such a lift exists, it is necessarily unique as we have a commutative diagram

$$\begin{tikzcd}
\mathcal{H}(\underline{\mathcal{M}\mathsf{fld}}_{/X}, \mathcal{V}) \arrow[r, dashed] \arrow[d] & \mathcal{H}(\underline{\mathcal{M}\mathsf{fld}}_{/Y}, \mathcal{V}) \arrow[r, "\simeq"] \arrow[d] & \mathbf{Alg}_{/Y}(\mathcal{V}) \arrow[d, hook] \\
\mathbf{Alg}_{\underline{\mathsf{Mfld}}_{/X}}(\mathcal{V}) \arrow[r] & \mathbf{Alg}_{\underline{\mathsf{Mfld}}_{/Y}}(\mathcal{V}) \arrow[r] & \mathbf{Alg}_{\underline{\mathsf{Disk}}_{/Y}}(\mathcal{V})
\end{tikzcd}$$

where the left vertical functor is fully-faithful by [AFT17a, Prop. 2.22]. Thus it suffices to prove that the induced $\underline{\mathsf{Disk}}_{/Y}$-algebra sends isotopy equivalences to equivalences in $\mathcal{V}$. Let $V \hookrightarrow V' \in \mathsf{Disk}_{/Y}$ be an isotopy equivalence. From property (i) of being homological, we have an equivalence $\mathcal{D}\mathsf{isk}_{/\alpha(V)} \simeq \mathcal{D}\mathsf{isk}_{/\alpha(V')}$. For $A \in \mathbf{Alg}_X(\mathcal{V})$, this induces a canonical equivalence $H(\alpha(V)) \simeq H(\alpha(V'))$ in $\mathcal{V}$, proving the result. ■

We now describe four different classes of maps between stratified spaces which induce homological morphisms, and hence allow us to transfer algebraic structures between them. We will denote these **homological maps**.

- **Open Embeddings:** As explained at the end of Definition 3.2, open embeddings induce morphisms of marked operads and in particular, we get an action $\mathbf{Aut}(X) \curvearrowright \mathcal{D}\mathsf{isk}_{/X}$. On π_0, this can be seen as acting through homological marked morphisms.

- **Refinements:** As explained in [AFT17a, Lem. 2.23], refinements of stratified spaces induce morphisms of marked operads, which are immidiately seen to be homological.
- **Weak Constructible Bundles:** Recall that a stratified map $\pi\colon X \to Y$ is a *weak constructible bundle* if for each stratum $Y_q \subset Y$, the corestriction $\pi_{|\pi^{-1}(Y_q)} : \pi^{-1}(Y_q) \to Y_q$ is a fiber bundle (see [AFT17b, Def. 3.6.1]). As explained in [AFT17a, Lem. 2.24 and Thm. 2.25], weak constructible bundles of stratified spaces induce homological morphisms of marked operads by taking preimages.
- **Closed Strata Inclusions:** For $Y \hookrightarrow X$ the inclusion of a closed union of strata, we have a canonical morphism of operads $\mathcal{D}\mathsf{isk}_{/X} \to \mathcal{D}\mathsf{isk}_{/Y}$ which "forgets" the algebraic data corresponding to the other strata. We can define this functor as follows. From [AFT17b, Prop. 8.2.5], we get a natural proper constructible bundle $\mathsf{Link}_Y(X) \xrightarrow{\pi} X$ inducing a cone construction

$$Y \twoheadleftarrow C(\pi) \dashrightarrow C(\pi)_{\text{unref.}} \hookrightarrow X$$

  where the left map is a constructilble bundle, the center one is a refinement and the right one is a open embedding. The construction of the right map can be seen as a stratified notion of a tubular neighborhood of Y in X. Note that the preimage of a basic in Y is basic in $C(\pi)$. Thus, one gets a homological marked morphism induced by $\underline{\mathsf{Disk}}_{/Y} \to \underline{\mathsf{Disk}}_{/C(\pi)} \to \underline{\mathsf{Disk}}_{/X}$.

  Note that the choice of tubular neighborhood is contractible here. More precisely, the neighbourhoods are given by considering the functorial "unzipping" construction (see [AFT17b, Sec. 7.2]) and picking an inward vector field which is non-vanishing on the link (the boundary of the unzipping procedure). The space of such vector fields is contractible. The construction can be seen as this space of vector field mapping to $\mathsf{Emb}(C(\pi)_{\text{unref.}}, Y)$ which itself maps to the space of operads morphisms of the type $\underline{\mathcal{D}\mathsf{isk}}_{/C(\pi)_{\text{unref.}}} \to \underline{\mathcal{D}\mathsf{isk}}_{/X}$. Thus any two choices of tubular neighborhoods will give rise to naturally equivalent functors $\mathcal{H}(\underline{\mathcal{M}\mathsf{fld}}_{/X}, \mathcal{V}) \to \mathcal{H}(\underline{\mathcal{M}\mathsf{fld}}_{/Y}, \mathcal{V})$.

The following result is the key technical tool of this paper on the factorization homology side.

Lemma 3.5 (Commutative Diagrams)

Let $\mathcal{V} \in \mathbf{Alg}_{\mathcal{E}_\infty}(\mathbf{Cat}_{(\infty,1)}^{\mathsf{geom}-\text{colim}})$. Any commutative diagram of homological maps which induces a commutative diagram of homological morphisms between the corresponding $\underline{\mathsf{Mfld}}_{/_}$ will also induce a commutative diagram of functors between the corresponding $\mathbf{Alg}_{/_}(\mathcal{V}) \simeq \mathcal{H}(\underline{\mathcal{M}\mathsf{fld}}_{/_}, \mathcal{V})$.

Proof: As homologicality is closed under composition (Lemma 3.3), the total composition is homological as well. This implies that there is a unique lift at the level of homology theories, and hence categories of algebras. The unicity of the lift precisely gives the commutativity of the diagram. ■

3.2 Handlebodies as Duality Datum

The goal of this subsection is to construct the stratified spaces which allow us to describe the "duality datum" of a k-morphism. We fix $n \geq 1$ throughout this section. We consider various stratifications of $\mathbb{R}^n$ which will "carry" the algebraic structure of the duality data, through factorization homology's pushforward (**??**) and refinement (**??**) constructions. We denote the coordinates of a point $x \in \mathbb{R}^n$ by $(x_1, \ldots, x_n)$. The reader should think of the ith-coordinate as the direction of composition of i-morphisms in an (∞, n)-category.

The first stratification is to be thought of as the stratification which "carries" the structure of an arbitrary k-morphism:

Definition 3.5 (Bar-Stratification)**.** For $0 \leq k \leq n$, we define $\mathbb{R}^n_{\overline{k}}$ to be the stratification of $\mathbb{R}^n$ given by the following strata:

- for $1 \leq i \leq k$, the strata given by $x_1 = x_2 = \ldots = x_{i-1} = 0$, and $\pm x_i > 0$,
- the stratum given by $x_1 = x_2 = \ldots = x_{k-1} = x_k = 0$.

This is a total of $2k + 1$ strata (by convention, when $k = 0$, there is a single stratum: $\mathbb{R}^n$ itself).

Remark 3.1 (Algebras over Bar-Stratifications)*.* If $k < n$, the data of an algebra $M \in \mathbf{Alg}_{/\mathbb{R}^n_{\overline{k}}}(\mathcal{V})$ is precisely the data of a k-morphism in the $(\infty, n+1)$-category $\mathbf{Mor}_n(\mathcal{V})$. If $k = n$, any such algebra gives a n-morphism in $\mathbf{Mor}_n(\mathcal{V})$, together with a choice of a point (an $\mathcal{E}_0$-structure).

The (co)domain and composition constructions can be seen as the following operations. For the rest of the exposition, we fix:

- a smooth function $\rho\colon \mathbb{R} \to [0,\infty)$ which is strictly decreasing on $(-\infty, 0]$, such that $\forall n \geq 0, \rho^{(n)}(0) = 0$ and $\forall x \geq 0, \rho(x) = 0$. For example, one may use $\rho(x) = e^{1/x}$ for all $x < 0$ and $\rho(x) = 0$ for $x \geq 0$.
- an increasing diffeomorphism $\phi\colon \mathbb{R} \to \mathbb{R}_{>0}$.

It is important to note that the choice of these functions does not matter (there is a contractible choice), although a precise statement of this fact is not necessary for our purposes.

Definition 3.6 ((Co)Domain)**.** For $1 \leq k \leq n$, we have an open embedding $\mathsf{dom}\colon \mathbb{R}^n_{\overline{k-1}} \hookrightarrow \mathbb{R}^n_{\overline{k}}$ which sends

$$x_k \mapsto \phi(x_k)$$

This induces an equivalence $\mathbb{R}^n_{\overline{k-1}} \simeq \left(\mathbb{R}^n_{\overline{k}}\right)_{x_k > 0}$. Similarly, one has $\mathsf{cod}\colon \mathbb{R}^n_{\overline{k-1}} \hookrightarrow \mathbb{R}^n_{\overline{k}}$ given by $x_k \mapsto -\phi(-x_k)$

Definition 3.7 (Composition)**.** For $1 \leq k \leq n$, we define $\mathbb{R}^n_{\overline{k \circ k}}$ to be the stratification of $\mathbb{R}^n$ given by the following strata:

- for $1 \leq i < k$, the strata given by $x_1 = x_2 = \ldots = x_{i-1} = 0$, and $\pm x_i > 0$,
- the strata given by $x_1 = x_2 = \ldots = x_{k-1} = 0$ and $x_k < -1$, $x_k = -1$, $-1 < x_k < 1$, $x_k = 1$, $1 < x_k$.

This is a total of $2k+3$ strata. We have a constructible bundle $\mathbb{R}^n_{\overline{k \circ k}} \overset{\circ}{\twoheadrightarrow} \mathbb{R}^n_{\overline{k}}$ which sends

$$x_k \mapsto x_k \left(\rho(1 - x_k^2) + \sum_{i=1}^{k-1} x_i^2 \right)$$

intuitively pinching the two $(n-k)$-dimensional strata.

The second stratification of interest is to be thought of as the stratification which "carries" the structure of some specific duality datum of an arbitrary k-morphism.

Definition 3.8 (Duality Data)**.** Let $d < n$ and a word $\sigma \in \{\epsilon, \eta\}^d$ of length d in the letters "ϵ, η" is called a d-duality datum. We denote $|\eta| = 1$ and $|\epsilon| = -1$.

Definition 3.9 (Handlebody-Stratification)**.** Let $1 + d \leq k + d \leq n$ and $\sigma = \sigma_{k+d-1} \ldots \sigma_k \in \{\epsilon, \eta\}^d$ be a d-duality datum. We define $\mathbb{R}^n_{\overline{k,\sigma}}$ to be the stratification of $\mathbb{R}^n$ given by the following strata:

- **Bar-strata:** for $1 \leq i \leq k-1$, the strata given by $x_1 = x_2 = \ldots = x_{i-1} = 0$, and $\pm x_i > 0$,
- **Handlebody-strata:** $x_1 = x_2 = \ldots = x_{k-1} = 0$, and
  (a) $x_{k+d} > \sum_{i=k}^{k+d-1} |\sigma_i| x_i^2$,
  (b) $x_{k+d} = \sum_{i=k}^{k+d-1} |\sigma_i| x_i^2$,
  (c) $x_{k+d} < \sum_{i=k}^{k+d-1} |\sigma_i| x_i^2$.

This is a total of $2k+1$ strata.

Note that the stratification above is isomorphic to the k-bar-stratification by sending x_{k+d} to $x_{k+d} - \sum_{i=k}^{k+d-1} |\sigma_i| x_i^2$, and by permuting the order of the coordinates. In particular, algebras over $\mathbb{R}^n_{\overline{k,\sigma}}$ are also (pointed) k-morphisms in $\mathbf{Mor}_n(\mathcal{V})$, although maybe in a less canonical way (as it depends on the choice of permutation).

Finally, the third stratification is to be thought of as the stratification which "carries" the understanding that σ-duality data of an arbitrary k-morphism is a $(k+d)$-morphism. Strangely enough, the author encourages the reader to skip this definition on a first reading, and to come back to it after Definition 3.12. Indeed, the stratification might seem quite complicated but it is simply the minimal one which makes the "full-pinch" map into a stratified one.

Definition 3.10 (Fully-Barred Handlebody-Stratification)**.** Let $1 + d \leq k + d \leq n$ and $\sigma = \sigma_{k+d-1} \ldots \sigma_k \in \{\epsilon, \eta\}^d$ be a d-duality data. We define $\mathbb{R}^n_{\overline{k,\overline{\sigma}}}$ to be the stratification of $\mathbb{R}^n$ given by the following strata:

- **Bar-strata:** for $1 \leq i \leq k-1$, the strata given by $x_1 = x_2 = \ldots = x_{i-1} = 0$, and $\pm x_i > 0$,

- **Handlebody-Bar-strata:** Write $\{|\sigma| = 1\} = \{k \leq i < k+d \mid |\sigma_i| = 1\}$, and similarly for $\{|\sigma| = -1\}$. For $k \leq i < k+d$, write $\Delta_i = |\sigma_i|\left(x_{k+d} - \sum_{j=i}^{k+d-1} |\sigma_i| x_i^2\right)$ with the convention $\Delta_{k+d} = -1$. For $k \leq j \leq k+d$, we have the following strata

  (a) the 2 strata given by $\pm x_j > 0$, $\Delta_j < 0$ and $x_u = 0$ for $u \in \{1, \ldots, j-1\}$,

  (b) for $k \leq i < j$, the 3 strata given by $x_u = 0$ for $u \in \{1, \ldots, k-1\} \sqcup \{|\sigma| = |\sigma_i|\} \cap \{k, \ldots, i-1\} \sqcup \{|\sigma| = -|\sigma_i|\} \cap \{k, \ldots, j-1\}$ and by either

     i. $\Delta_i > 0$, or

     ii. $\Delta_i = 0$ and $\pm x_i > 0$.

  Finally, we have the stratum given by $x_1 = \ldots = x_{k+d} = 0$.

Note that we have an obvious refinement map $\mathbb{R}^n_{\overline{k},\overline{\sigma}} \dashrightarrow \mathbb{R}^n_{\overline{k},\sigma}$.

We now define some "pinch maps" which will help formalize the idea that the σ-duality data of an arbitrary k-morphism is a $(k+d)$-morphism.

Definition 3.11 (Last Coordinate Pinch)**.** As above, let $1+d \leq k+d \leq n$ and $\sigma = \sigma_{k+d-1} \ldots \sigma_k \in \{\epsilon, \eta\}^d$ be a d-duality data. We define a stratified map $\pi_k \colon \mathbb{R}^n_{\overline{k},\overline{\sigma}} \to \mathbb{R}^n_{\overline{k+1},\overline{\sigma_{k+d-1}\ldots\sigma_{k+1}}}$ called the "pinch map" by the following:

- it leaves unchanged all coordinates except the kth-one,
- it sends the kth-coordinate x_k to

$$x_k \longmapsto x_k\left(\rho\left(|\sigma_k|\left(x_{k+d} - \sum_{i=k}^{k+d-1} |\sigma_i| x_i^2\right)\right) + \sum_{i=1}^{k-1} x_i^2\right)$$

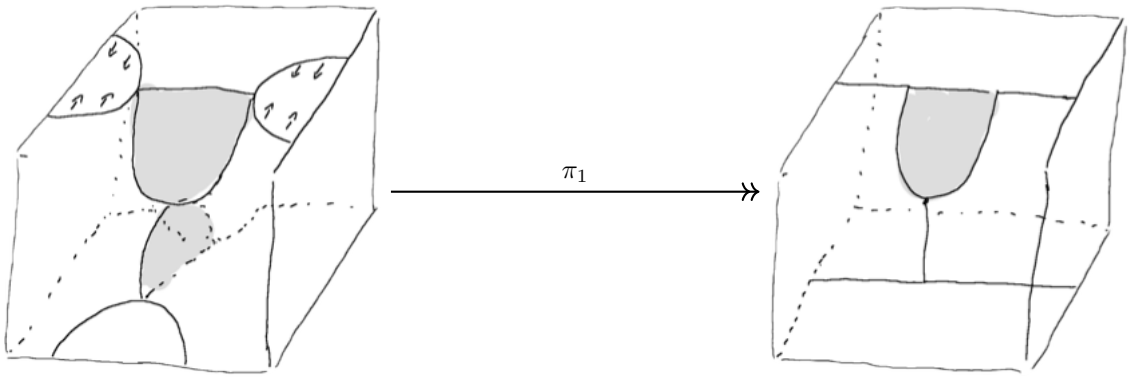

Figure 1: Pinching $\mathbb{R}^3_{\overline{1},\overline{-1,1}} \xrightarrow{\pi_1} \mathbb{R}^3_{\overline{1},\overline{1}}$. The small arrows indicate the collapsing that occurs "inside" the handle.

While we only need the easer notion of weak constructible bundle for our purposes, one can easily see that these maps are constructible bundle.

Lemma 3.6

As above, let $1+d \leq k+d \leq n$ and $\sigma = \sigma_{k+d-1} \ldots \sigma_k \in \{\epsilon, \eta\}^d$ be a d-duality data. The pinch

$$\mathbb{R}^n_{\overline{k},\overline{\sigma}} \xrightarrow{\pi_k} \mathbb{R}^n_{\overline{k+1},\overline{\sigma_{k+d-1}\ldots\sigma_{k+1}}}$$

is a (weak) constructible bundles, i.e. a stratified fiber bundle when corestricted to each strata of the codomain.

Proof: Consider a stratum of the codomain $\mathbb{R}^n_{\overline{k+1},\overline{\sigma_{k+d-1}\ldots\sigma_{k+1}}}$. We have to check that the preimage of the pinch map is a stratified fiber bundle over it. To simplify notations, we write $\overline{x}_k$ for the new k^{th}-coordinate after applying the pinch map. We denote the strata using a "path" notation indcating the order of the choices.

- **Bar-strata:** The bar-strata given by $i.\pm$ are unchanged for $i \leq k-1$. The preimage of the bar-strata $k.\pm$ are given by the handlebody-bar-strata $k.a.\pm$. For both these cases, the pinch map is diffeomorphic.

- **Handlebody-Bar-Strata:** Let $k+1 \leq j \leq k+d$. If $\sigma_k = \sigma_j$, the preimage of the handlebody-bar-strata $j.a.\pm$ are given by the handlebody-bar-strata $j.a.\pm$, and the pinch is diffeomorphic. If $\sigma_k \neq \sigma_j$, the preimage of the handlebody-bar-strata $j.a.\pm$ are given by the union of the three handlebody-bar-strata $j.k._$, where the pinch is pointwise given by a map equivalent to $[-1,1] \to \{0\}$. On these strata, one can check that the pinch is a fiber bundle. For $k+1 \leq i \leq j$, the preimage of the handlebody-bar-strata $j.i._$ are given by the corresponding handlebody-bar-strata $j.i._$, where the pinch is diffeomorphic.
- **Bottom-Bar-Stratum:** Corestricted to the stratum given by $x_1 = \ldots = x_{k+d} = 0$, the pinch map is the identity, and hence a fiber bundle. ■

In particular, we can compose these pinch maps to get the corollary bellow.

Definition 3.12 (Full Pinch)**.** As above, let $1+d \leq k+d \leq n$ and $\sigma = \sigma_{k+d-1} \ldots \sigma_k \in \{\epsilon, \eta\}^d$ be a d-duality data. We define the full pinch as the compositions

$$\mathbb{R}^n_{\overline{k},\overline{\sigma}} \twoheadrightarrow \mathbb{R}^n_{\overline{k+1},\overline{\sigma_{k+d-1}\ldots\sigma_{k+1}}} \twoheadrightarrow \ldots \twoheadrightarrow \mathbb{R}^n_{\overline{k+d-1},\overline{\sigma_{k+d-1}}} \twoheadrightarrow \mathbb{R}^n_{\overline{k+d},\overline{\emptyset}} = \mathbb{R}^n_{\overline{k+d}}$$

We define $\sigma\colon \mathbf{Alg}_{/\mathbb{R}^n_{\overline{k}}}(\mathcal{V}) \to \mathbf{Alg}_{/\mathbb{R}^n_{\overline{k+d}}}(\mathcal{V})$ to be the composition of the functors induced by the homological maps (see subsection 3.1)

$$\mathbb{R}^n_{\overline{k}} \xrightarrow{\sim} \mathbb{R}^n_{\overline{k},\sigma} \dashleftarrow \mathbb{R}^n_{\overline{k},\overline{\sigma}} \twoheadrightarrow \mathbb{R}^n_{\overline{k+d}}$$

Corollary 3.7

As above, let $1+d \leq k+d \leq n$ and $\sigma \in \{\epsilon, \eta\}^d$ be a d-duality data. The full pinch $\mathbb{R}^n_{\overline{k},\overline{\sigma}} \to \mathbb{R}^n_{\overline{k+d}}$ is a (weak) constructible bundle, i.e. a fiber bundle when restricted to each strata of the codomain.

Proof: This is a direct consequence of [AFR18b, Prop. 6.1.8]. ■

We briefly check that, for $d = 1$, the notation $\eta(f)$ and $\epsilon(f)$ introduced above do indeed exhibit an adjunction in the higher Morita category.

Lemma 3.8 (Duality Data Forms Adjunctions)

Let $k < n$ and $f \in \mathbf{Alg}_{/\mathbb{R}^n_{\overline{k}}}(\mathcal{V})$ be a k-morphism in $\mathbf{Mor}_n(\mathcal{V})$. Let f° be the k-morphism induced by $x_k \mapsto -x_k$ on $\mathbb{R}^n_{\overline{k}}$. Then, the $(k+1)$-morphisms $\eta(f)$ and $\epsilon(f)$ are respectively the unit and counit of an adjunction $f \dashv f^\circ$ in $\mathbf{Mor}_n(\mathcal{V})$.

Proof: By inspection, $\mathsf{cod}(\eta(f)) \simeq f^\circ \circ f$ and $\mathsf{dom}(\epsilon(f)) \simeq f \circ f^\circ$ using Definition 3.6 and Definition 3.7. The other domain and codomain can similalry be identitfied. The triangle identities then follow from the usual graphical calculus for adjunctions. ■

Note that, by construction, $(f^\circ)^\circ \simeq f$ and $\eta(f^\circ) \simeq \epsilon(f)^\circ$. We then have a direct corollary of Proposition 2.15.

Corollary 3.9 (Units in Morita)

Let $k < n$ and $f \in \mathbf{Alg}_{/\mathbb{R}^n_{\overline{k}}}(\mathcal{V})$ be a k-morphism in $\mathbf{Mor}_n(\mathcal{V})$. We have a lift $c_k \twoheadrightarrow \mathsf{Free}^{\{k\}}(c_k) \to \mathbf{Mor}_n(\mathcal{V})$ where the all induced (co)units are given by either of the following four algebras: $\eta(f)$, $\epsilon(f)$, $\eta(f^\circ)$, $\epsilon(f^\circ)$.

We now prove an associativity lemma that can be understood as "$\sigma(\eta(M)) \simeq \sigma\eta(M)$" (resp. for ϵ). This lemma is key to our inductive process in the proof of Theorem 4.2.

Lemma 3.10 (Left Associativity)

Let $1+d \leq k+d \leq n$ with $d \geq 2$ and let $\sigma = \sigma_{k+d-1} \ldots \sigma_k \in \{\epsilon, \eta\}^d$ be a d-duality data. Define $\sigma' = \sigma_{k+d-1} \ldots \sigma_{k+2} \in \{\epsilon, \eta\}^{d-1}$. We have a commutative diagram

$$\begin{tikzcd}
 & \mathbf{Alg}_{/\mathbb{R}^n_{\overline{k+1}}}(\mathcal{V}) \arrow[dr, "\sigma'"] & \\
\mathbf{Alg}_{/\mathbb{R}^n_{\overline{k}}}(\mathcal{V}) \arrow[ur, "\sigma_{k+1}"] \arrow[rr, "\sigma"'] & & \mathbf{Alg}_{/\mathbb{R}^n_{\overline{k+d}}}(\mathcal{V})
\end{tikzcd}$$

Proof: One can define a "partially-bar" handlebody stratification $\mathbb{R}^n_{\overline{k},\overline{\sigma_k},\sigma'}$, refining $\mathbb{R}^n_{\overline{k},\sigma}$ and minimal such that the pinch map $\mathbb{R}^n_{\overline{k},\overline{\sigma_k},\sigma'} \xrightarrow{\pi_k} \mathbb{R}^n_{\overline{k+1},\sigma'}$ is a map of stratified spaces. We then have a commutative diagram of homological maps between stratified spaces

$$\begin{tikzcd}
\mathbb{R}^n_{\overline{k}} \arrow[r, dash, "\sim"] \arrow[d, dash, "\wr"] & \mathbb{R}^n_{\overline{k},\sigma_{k+1}} & \mathbb{R}^n_{\overline{k},\overline{\sigma_{k+1}}} \arrow[l, dashed] \arrow[r, two heads] \arrow[d, dash, "\wr"] & \mathbb{R}^n_{\overline{k+1}} \arrow[d, dash, "\wr"] \\
\mathbb{R}^n_{\overline{k},\sigma} \arrow[urr, dashed] & & \mathbb{R}^n_{\overline{k},\overline{\sigma_{k+1}},\sigma'} \arrow[ll, dashed] \arrow[r, two heads] & \mathbb{R}^n_{\overline{k+1},\sigma'} \\
\mathbb{R}^n_{\overline{k},\overline{\sigma}} \arrow[u, dashed] \arrow[urrr, dashed] \arrow[rrr, two heads] & & & \mathbb{R}^n_{\overline{k+1},\overline{\sigma'}} \arrow[u, dashed] \arrow[r, two heads] & \mathbb{R}^n_{\overline{k+d}}
\end{tikzcd}$$

Clearly, this diagram satisifes the condition of Lemma 3.5 and hence induces a commutative diagram of functors between the corresponding categories of algebras. The outer top composition induces $\sigma' \circ \sigma_{k+1}$, while the outer bottom composition induces σ. ■

3.3 Dualizability of Iterated Bimodules

This subsection is solely devoted to the result bellow. We mimick here an argument from [Lur17, Prop. 4.6.2.13] to identify dualizable iterated bimodules, using a similar idea to [Ram24, Cor. 3.8].

Proposition 3.11 (Dualizability for Iterated Bimodules)

Let M be a $\mathbb{R}^n_{\overline{n}}$-algebra in $\mathcal{V}$, seen as an n-morphism in $\mathbf{Mor}_n(\mathcal{V})$. Then, we have

- *M is a left adjoint in $\mathbf{Mor}_n(\mathcal{V})$ if and only if the induced $\mathbb{R}_{\geq 0}$-algebra, seen as a 1-morphism to the trivial algebra in $\mathbf{Mor}_1(\mathcal{V})$, is a left adjoint,*
- *M is a right adjoint in $\mathbf{Mor}_n(\mathcal{V})$ if and only if the induced $\mathbb{R}_{\leq 0}$-algebra, seen as a 1-morphism from the trivial algebra in $\mathbf{Mor}_1(\mathcal{V})$, is a right adjoint.*

Proof: The second statement follows from the first by the "n-op" operation $x_n \mapsto -x_n$. Consider the following composition of constructible bundles $\mathbb{R}^n_{\overline{n}} \twoheadrightarrow \left(\mathbb{R}^2_{\overline{2}}\right)_{r\geq 0} \twoheadrightarrow \mathbb{R}^1_{\overline{1}}$, where the coordinates of the middle copy of $\mathbb{R}^2$ are denoted (x_n, r), defined by

$$(x_1, \ldots, x_n) \longmapsto \left(x_n, \sqrt{\sum_{i=1}^{n-1} x_i^2}\right) \longmapsto x_n$$

Using the iterated construction of the Morita category [Hau23, Cor. 7.6], or the factorization homology construction of composition in $\mathbf{Mor}_n(\mathcal{V})$, we see that M is a left adjoint in $\mathbf{Mor}_n(\mathcal{V})$ if and only if it is a left adjoint as a 1-morphism in $\mathbf{Mor}_1(\mathbf{Mod}_W(\mathcal{V}))$ where W is the $\mathcal{E}_2$-algebra supported on $r > 0$ in the pushforward $(x_n, r)^*(M)$. By [Lur17, Prop. 4.6.2.13], this is equivalent to the induced $\mathbb{R}_{\geq 0}$-algebra being a left adjoint in $\mathbf{Mor}_1(\mathbf{Mod}_W(\mathcal{V}))$. It remains to show that this is equivalent to being a left adjoint in $\mathbf{Mor}_1(\mathcal{V})$. One direction comes from stability of adjunctions under composition, so it suffices to show that being a left adjoint in $\mathbf{Mor}_1(\mathcal{V})$ implies being a left adjoint in $\mathbf{Mor}_1(\mathbf{Mod}_W(\mathcal{V}))$. The first part of the argument relies on a similar idea that the proof of [Lur17, Prop. 4.6.2.13], while the second part is a consequence of (a harmless generalization) of the argument in [Ram24, Cor. 3.8].

Let $\mathcal{W} = \mathbf{Mod}_W(\mathcal{V})$ and let B denote the induced $\mathbb{R}_{\geq 0}$-algebra. Note that $\mathbf{Mod}_B(\mathcal{V}) \simeq \mathbf{Mod}_B(\mathcal{W})$ by , so we can abuse notation and denote both categories by $\mathbf{Mod}_B$. Consider the following lifitng problem: **Pablo:** to detail

$$\begin{tikzcd}
& \mathcal{W} \arrow[dl, bend right, "W\otimes -"'] \arrow[dr, bend left, "B\otimes -"] & \\
\mathcal{V} \arrow[ur, bend left, "\vdash"] \arrow[rr, bend left, "B^\vee\otimes -"] & & \mathbf{Mod}_B \arrow[ul, dashed, bend right, "\dashv"'] \arrow[ll, bend left, "\perp"] \arrow[ll, bend right, "B\otimes -"']
\end{tikzcd} \tag{3.1}$$

where we'd like to show the dotted left adjoint exists. We can consider the full subcategory $\mathcal{W}' \hookrightarrow \mathcal{W}$ on the objects w such that $\mathcal{W}(w, B\otimes -)\colon \mathbf{Mod}_B \to \mathcal{S}$ is corepresentable. Clearly, $\mathcal{W}'$ contains the free W-algebras (the image of the functor $\mathcal{V} \to \mathcal{W}$) and is closed under geometric realizations. Hence, $\mathcal{W}' = \mathcal{W}$ and the left adjoint exists.

Automatically, this left adjoint preserves colimits. To show that it is given by tensoring with some lift $B^\vee \in \mathcal{W}$, it suffices to check that this adjoint is a $\mathcal{W}$-module map. This follows from the argument in [Ram24, Cor. 3.8]. As a left adjoint to a module map, it is automatically an oplax-$\mathcal{W}$ map. Because $\mathcal{W}$ is generated

under colimits by its unit W, it is in particular generated under colimits by dualizable objects. The "trace map" construction of [Ram24, Prop. 3.4] then shows that the oplax structure maps are equivalences. ■

In particular, we immidiately get the following corollary in the category of spectra.

Corollary 3.12 (Dualizability for Iterated Bimodules in Spectra)

Suppose $\mathcal{V} = \mathcal{S}p$. Let M be a $\mathbb{R}^n_{\overline{n}}$-algebra in $\mathcal{V}$, seen as an n-morphism in $\mathbf{Mor}_n(\mathcal{V})$. Then, we have

- *M is a left adjoint in $\mathbf{Mor}_n(\mathcal{V})$ if and only if it is compact as a module over the induced $\mathbb{R}_{>0}$-algebra, if and only if it is perfect as a module over the induced $\mathbb{R}_{>0}$-algebra,*
- *M is a right adjoint in $\mathbf{Mor}_n(\mathcal{V})$ if and only if it is compact as a module over the induced $\mathbb{R}_{<0}$-algebra, if and only if it is perfect as a module over the induced $\mathbb{R}_{<0}$-algebra,*

Proof: This is a direct consequence of [Lur17, 7.2.4.4]. ■

3.4 Centers for Dualizable Algebras

We recall here the content of [Lur17, Sec. 5.3.1] and detail how it applies to our setting.

Lemma 3.13 (Units as Maps to Centers)

Let $(\mathcal{A}, \mathcal{M}) \in \mathbf{Alg}_{\mathbb{R}_{\geq 0}}(\mathbf{Cat}^{\mathsf{geom}-\mathrm{colim}}_{(\infty,1)})$ and (A, M) a $\mathcal{D}\mathsf{isk}/\mathbb{R}_{\geq 0}$-algebra in $(\mathcal{A}, \mathcal{M})$. Suppose that we can extend the diagram bellow with the following dashed right adjoints.

$$\begin{tikzcd}
 & \mathbf{Mod}_A(\mathcal{A}) \arrow[dl, bend right, "-\otimes A"'] \arrow[dr, bend left, "-\otimes_A M"] \arrow[dl, bend left, "\dashv"] \arrow[dr, dashed, bend right, "\dashv"'] & \\
\mathcal{A} \arrow[rr, bend left, "-\otimes M", "\perp"'] & & \mathcal{M} \arrow[ll, dashed, bend left, "\hom_{\mathcal{M}}(M,-)"]
\end{tikzcd} \tag{3.2}$$

Then, when evaluating at $\mathbb{1}_{\mathcal{A}}$, the monad of the bottom adjuntion gives the center of M (seen as an object in $\mathcal{A}$) and the induced map of monads gives the universal $A \to \hom_{\mathcal{M}}(M, M)$ corresponding to the action of $A \curvearrowright M$ (seen as a morphism in $\mathcal{A}$).

Note that we only need and make this claim when seeing the map $A \to \hom_{\mathcal{M}}(M, M)$ as a map in $\mathcal{A}$. It is also true as a map in $\mathbf{Alg}_{\mathcal{E}_1}(\mathcal{A})$, using the induced $\mathcal{E}_1$-algebra structure on the map of monads, but we will not need this fact here.

Proof: By definition, $\hom(M, M)$ is an endormorphism object for M [Lur17, Def. 4.2.1.28], and hence a center for M [Lur17, Lem. 5.3.1.11]. By the universal property of morphism objects, to identify maps $A \to \hom_{\mathcal{M}}(M, M)$, it suffices to identify the induced maps $A \otimes M \to M$. It is then given by construction that the map of monads induces the correct action map on M. ■

Remark 3.2 (Lift of the Action of A on the Dual). One can actually prove the top dashed arrow exists if and only if the bottom dashed arrow exists. This is essentially contained in [Lur17, Sec. 5.3.1]. One way to see this is to observe that the left hand side adjunction is monadic and following the ideas of [Lur17, Sec. 4.7.3], we have: a (left) action of A on M gives a (right) action of the monad $-\otimes A$ on $-\otimes M$, which in turn is equivalent to a (left) action of the monad $-\otimes A$ on $\hom_{\mathcal{M}}(M, -)$, which in turn is equivalent to the existence of the dotted lift.

To complete the picture, we propose the following approach to re-prove the results of [Lur17, Sec. 5.3.1]. Now suppose $(\mathcal{A}, \mathcal{M}) \in \mathbf{Alg}_{\mathbb{R}_{\leq 0} \times \mathbb{R}^n}(\mathbf{Cat}^{\mathsf{geom}-\mathrm{colim}}_{(\infty,1)})$ and (A, M) a $\mathcal{D}\mathsf{isk}_{/\mathbb{R}_{\leq 0} \times \mathbb{R}^n}$-algebra in $(\mathcal{A}, \mathcal{M})$. Consider a similar diagram, but replacing $\mathcal{M}$ with $\mathbf{Mod}^{\bullet}_M(\mathcal{M})$, and assume again the existence of the dashed right adjoints. All the left adjoint functors are $\mathcal{E}_n$-monoidal. Then, as can also be seen by [HHLN24, Cor. C], all the right

adjoint functors are canoncially lax-$\mathcal{E}_n$-monoidal and the diagram induces

$$\begin{tikzcd}
& \mathbf{Alg}_{\mathcal{E}_n}(\mathbf{Mod}_A(\mathcal{A})) \arrow[dl, shift right, "-\otimes A"'] \arrow[dr, shift left, "-\otimes_A M"] \arrow[dl, phantom, shift left=3, "\curlyvee"] \arrow[dr, phantom, shift right=3, "\curlyvee"] & \\
\mathbf{Alg}_{\mathcal{E}_n}(\mathcal{A}) \arrow[ur, shift right] \arrow[rr, shift left, "-\otimes M"] \arrow[rr, phantom, "\perp"] & & \mathbf{Alg}_{\mathcal{E}_n}(\mathbf{Mod}_M^{\bullet}(\mathcal{M})) \arrow[ul, shift left] \arrow[ul, dashed, shift right, "\hom_M^{\bullet}(M,-)"'] \arrow[r, "\sim"] & \mathbf{Alg}_{\mathcal{E}_n}(\mathcal{M})_{M\backslash}
\end{tikzcd}$$

Informally, the induced monad of the bottom adjunction, evaluated at the identity $\mathbb{1}_{\mathcal{A}}$, upgrades to $\mathbf{Alg}_{\mathcal{E}_1}(\mathbf{Alg}_{\mathcal{E}_n}(\mathcal{A})) \simeq \mathbf{Alg}_{\mathcal{E}_{n+1}}(\mathcal{A})$ and can be identified with the $\mathcal{E}_n$-center of M. The induced map of monad, evalutated at the identity $\mathbb{1}_{\mathcal{A}}$, can be identified with the universal $\mathcal{E}_{n+1}$-map to the center $A \to \hom_M^{\bullet}(M, M)$. Thankfully, we do not need here to access any of the $\mathcal{E}_{n+1}$-monoidal structures of these maps, so the following result suffices for our purposes.

Lemma 3.14 (Units as Maps to $\mathcal{E}_n$-Centers)

As above, let $(\mathcal{A}, \mathcal{M}) \in \mathbf{Alg}_{\mathbb{R}_{\leq 0}\times\mathbb{R}^n}(\mathbf{Cat}_{(\infty,1)}^{\xrightarrow{\mathsf{geom}-\mathrm{colim}}})$ and (A, M) a $\mathcal{D}\mathsf{isk}_{/\mathbb{R}_{\leq 0}\times\mathbb{R}^n}$-algebra in $(\mathcal{A}, \mathcal{M})$. Suppose that we can extend the diagram bellow with the following dashed right adjoints.

$$\begin{tikzcd}
& \mathbf{Mod}_A(\mathcal{A}) \arrow[dl, shift right, "-\otimes A"'] \arrow[dr, shift left, "-\otimes_A M"] \arrow[dl, phantom, shift left=3, "\curlyvee"] \arrow[dr, phantom, shift right=3, "\curlyvee"] & \\
\mathcal{A} \arrow[ur, shift right] \arrow[rr, shift left, "-\otimes M"] \arrow[rr, phantom, "\perp"] & & \mathbf{Mod}_M^{\bullet}(\mathcal{M}) \arrow[ul, shift left] \arrow[ul, dashed, shift right, "\hom_M^{\bullet}(M,-)"']
\end{tikzcd} \tag{3.3}$$

Then, when evaluating at $\mathbb{1}_{\mathcal{A}}$, the monad of the bottom adjuntion gives the $\mathcal{E}_n$-center of M (seen as an object in $\mathcal{A}$) and the induced map of monads gives the universal $A \to \hom_{\mathcal{M}}(M, M)$ corresponding to the action of $A \curvearrowright M$ (seen as a morphism in $\mathcal{A}$).

Proof: This is an immidiate consequence of Lemma 3.13 and of the proof of [Lur17, Thm. 5.3.1.30]. ■

In our case, the $-\otimes_A M$ functor on the top right side of the diagram can be defined explicity with factorization homology as follows:

Remark 3.3. Let $(A, M, B) \in \mathbf{Alg}_{/\mathbb{R}^{n+1}_{\overline{1}}}(\mathcal{V})$. We can consider $(\mathcal{A}, \mathcal{M}) = (\mathcal{V}, \mathbf{Mod}_B(\mathcal{V})$ above, which canonically inherits the correct algebra structure via the collapsing of the $x_1 \geq 0$ region $\mathbb{R}^{n+1}_{\overline{1}} \twoheadrightarrow \mathbb{R}_{\leq 0} \times \mathbb{R}^n$. The functor $\mathbf{Mod}_A(\mathcal{V}) \xrightarrow{-\otimes_A M} \mathbf{Mod}_M^{\bullet}(\mathbf{Mod}_B(\mathcal{V}))$ above is then given by the bimodule action induced by

$$\mathbb{R}^{n+1}_{\overline{1}} \xleftarrow{\text{cl.}} (\mathbb{R}_{\leq 0} \times \{0\}^n) \bigsqcup_{\{0\}^{n+1}} (\mathbb{R}_{\geq 0} \times \mathbb{R}^n) \twoheadrightarrow \mathbb{R}_{\overline{1}}$$

where the first map is a closed inclusion, and the second map is the constructible bundle induced by taking the radius in the $x_1 > 0$ region.

Using this, we can now prove the following:

Proposition 3.15 (Identifying Units with Universal Maps to Centers)

Let $k + d = n$ and $\sigma = \sigma_{k+d-1} \dots \sigma_k \in \{\epsilon, \eta\}^d$ be a d-duality data. Let $a = |\{k \leq i < k + d \mid \sigma_i = \epsilon\}|$ and $b = d - a$. Let M be a $\mathbb{R}^n_{\overline{k}}$-algebra in $\mathcal{V}$, seen as an k-morphism in $\mathbf{Mor}_n(\mathcal{V})$. Let A, B be respectively the induced $\mathbb{R}_{<0}$ and $\mathbb{R}_{>0}$-algebras, as in Proposition 3.11, applied to M seen as a $\mathbb{R}^k_{\overline{k}}$-algebra. Suppose that $\sigma(M)$, seen as a n-morphism in $\mathbf{Mor}_n(\mathcal{V})$, has a right adjoint $\sigma(M)^\vee$. Then, the unit of the adjunction can be identified with the image in $\mathcal{V}$ of the canonical $\mathcal{E}_{b+1}$-map

$$\int_{(S^{a-1}, D^a)} (M, A) \longrightarrow \mathbf{HC}^B_{\mathcal{E}_b}(M)$$

where $\mathbf{HC}^B_{\mathcal{E}_b}(M)$ is the $\mathcal{E}_b$-center with $(\mathcal{A}, \mathcal{M}) = (\mathcal{V}, \mathbf{Mod}_B(\mathcal{V}))$ above, and where the action is induced by

the homological maps (see subsection 3.1)

$$\begin{tikzcd}
 & \mathbb{R}^n_{\overline{k}} \\
(\{0\}^{a+1}\times\mathbb{R}^b \subset \{0\}\times\mathbb{R}^d \subset \mathbb{R}_{\geq 0}\times\mathbb{R}^d)\bigsqcup_{\{0\}^{a+1}\times\mathbb{R}^b}(\mathbb{R}_{\leq 0}\times\{0\}^a\times\mathbb{R}^b) \arrow[r, two heads] \arrow[ur, hook, "\mathrm{cl}"] & \mathbb{R}^{b+1}_{\overline{1}}
\end{tikzcd}$$

and where the top inclusion is induced by the inclusion of the last $d+1$ coordinates $\mathbb{R}^{d+1}_{\overline{1}} \overset{\mathrm{cl.}}{\hookrightarrow} \mathbb{R}^n_{\overline{k}}$.

Proof: Let $\overline{A}$ and $\overline{B}$ be the induced $\mathbb{R}_{\geq 0}$ and $\mathbb{R}_{\leq 0}$-algebras, as in Proposition 3.11, applied to $\sigma(M)$. One can dualize the diagram (3.1) and use it to extend the diagram (3.2) to the following:

$$\begin{tikzcd}
\mathbf{Mod}_{\overline{A}} \arrow[d, bend right, "-\otimes\overline{A}"'] \arrow[dr, bend left, "-\otimes_{\overline{A}}M"] \arrow[dr, phantom, "\succ"] & \\
\mathcal{W} \arrow[u, bend right, "\dashv"'] \arrow[r, bend left, "-\otimes M"] \arrow[r, phantom, "\perp"] \arrow[d, bend right, "-\otimes W"'] \arrow[d, phantom, "\dashv"] & \mathbf{Mod}_{\overline{B}} \arrow[ul, bend left, "\hom_{\mathbf{Mod}_{\overline{B}}}(M,-)"] \arrow[dl, bend left, "\hom_{\mathbf{Mod}_{\overline{B}}}(M,-)"] \arrow[dl, phantom, "\succ"] \\
\mathcal{V} \arrow[ur, bend left, "-\otimes M"] &
\end{tikzcd}$$

Under the downward forgetful functor $\mathcal{W} \to \mathcal{V}$, the unit of the top adjunction evaluated at $\overline{A}$ is equivalent to the induced map of monad of the top triangle, evaluated at $\mathbb{1}_{\mathcal{W}}$, which in turn is equivalent to the induced map of monad of the outer triangle, evaluated at $\mathbb{1}_{\mathcal{V}}$. By the above discussion, this unit can be identified with the canonical map to some center.

It now suffices to identify the functor $\mathbf{Mod}_{\overline{A}} \xrightarrow{-\otimes_{\overline{A}}M} \mathbf{Mod}_{\overline{B}}$ with the one induced by the action described in Remark 3.3 (where the "n" there is our "b"). By constuction, the functor $-\otimes_{\overline{A}} M$ above corresponds to the bimodule structure induced by the restriction to the last coordinate $\mathbb{R}_{\overline{1}} \hookrightarrow \mathbb{R}^n_{\overline{n}}$:

$$\begin{tikzcd}
 & & X \arrow[r, two heads] \arrow[d, hook, "\mathrm{cl}"] \arrow[dr, phantom, "\lrcorner", very near start] & \mathbb{R}_{\overline{1}} \arrow[d, hook, "\mathrm{cl}"] \\
\mathbb{R}^n_{\overline{k}} & \mathbb{R}^n_{\overline{k},\sigma} \arrow[l, "\sim"'] & \mathbb{R}^n_{\overline{k},\overline{\sigma}} \arrow[l, "\mathrm{ref}"'] \arrow[r, two heads] & \mathbb{R}^n_{\overline{n}}
\end{tikzcd}$$

By Definition 3.10, the top map $X \twoheadrightarrow \mathbb{R}_{\overline{1}}$ factorises as $X \simeq \mathbb{R}_{\leq 0}\times\mathbb{R}^a\times\{0\}^b \bigsqcup_{\{0\}^{d+1}} \mathbb{R}_{\geq 0}\times\{0\}^a\times\mathbb{R}^b \twoheadrightarrow \mathbb{R}_{\overline{1}}$, where the second map is the radius on each factor. The equivalence is given by some permutation of coordinates. We then have the following diagram (see the figure bellow for a visual interpretation):

$$\begin{tikzcd}
 & & & & \mathbb{R}^{b+1}_{\overline{1}} \\
 & Y \arrow[urrrr, two heads] & & & (\mathbb{R}_{\leq 0}\times\{0\}^b)\bigsqcup_{\{0\}^{b+1}}(\mathbb{R}_{\geq 0}\times\mathbb{R}^b) \arrow[u, hook, "\mathrm{cl}"'] \arrow[d, two heads] \\
\mathbb{R}^d_{\overline{1}} \arrow[d, hook] & \mathbb{R}_{\leq 0}\times\mathbb{R}^a\times\{0\}^b\bigsqcup_{\{0\}^{d+1}}\mathbb{R}_{\geq 0}\times\{0\}^a\times\mathbb{R}^b \arrow[l, hook'] \arrow[u, hook, "\mathrm{cl}"] \arrow[urr, two heads] & \sim & X \arrow[r, two heads] \arrow[d, hook, "\mathrm{cl}"] & \mathbb{R}_{\overline{1}} \arrow[d, hook, "\mathrm{cl}"] \\
\mathbb{R}^n_{\overline{k}} & \mathbb{R}^n_{\overline{k},\sigma} \arrow[l, "\sim"'] & & \mathbb{R}^n_{\overline{k},\overline{\sigma}} \arrow[ll, "\mathrm{ref}"'] \arrow[r, two heads] & \mathbb{R}^n_{\overline{n}}
\end{tikzcd} \tag{3.4}$$

where $Y = (\{0\}^{a+1}\times\mathbb{R}^b \subset \{0\}\times\mathbb{R}^d \subset \mathbb{R}_{\geq 0}\times\mathbb{R}^d)\bigsqcup_{\{0\}^{a+1}\times\mathbb{R}^b}(\mathbb{R}_{\leq 0}\times\{0\}^a\times\mathbb{R}^b)$ is the intermediate space in the statement of the result. By pulling back vector fields along the various weak constructible bundles, the diagram can be seen to satisfy the condition of Lemma 3.5, resulting in a commutative diagram of functors between categories of algebra. The left/top side corresponds to the action described in the statement. The right side corresponds to the construction of Remark 3.3. The bottom corresponds to the original description coming from Definition 3.10. ■

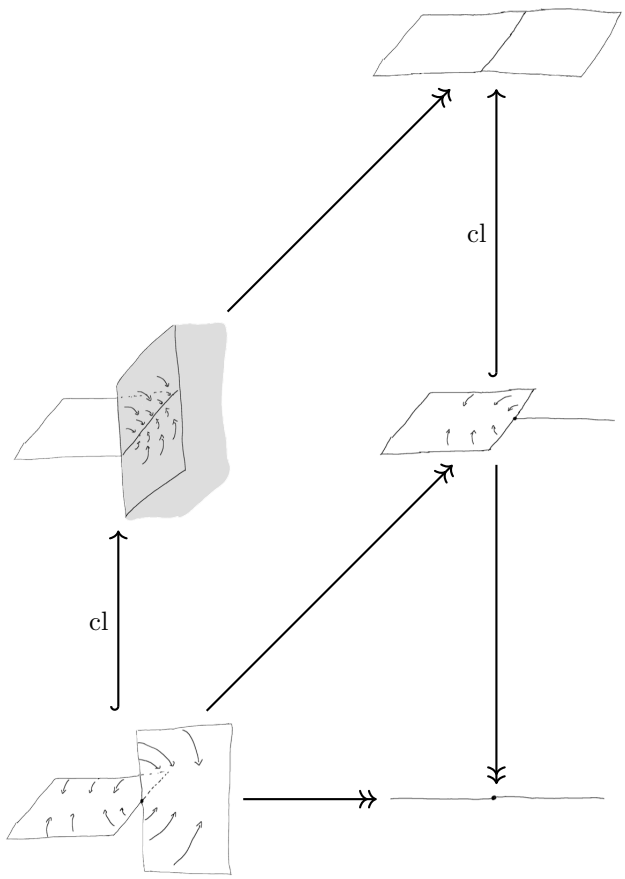

Figure 2: Visualization of the diagram (3.4).

4 Proof of the Criteria

In this final section, we prove our main Theorems (Theorem 4.2 and Theorem 4.5). The core of the argument is inductive: as proved in section 2, to show that some object is fully-dualizable (resp. invertible), knowing that it already is dualizable, it suffices to show its (co)units are fully-dualizable (resp. invertible). The tools developped in section 3 then allow to precisely descibe what happens when one iterates the "$f \mapsto$ (co)units of f" construction in the higher Morita categories. Combining both elements, we immidiately get our criteria.

4.1 Theorem A

Before heading into the proof, we establish a small well-known lemma. The idea behind this lemma is that the data exhibiting morphisms as higher full-adjoints is highly redundant. This property is sometimes known as ambidexterity in the literature.

Lemma 4.1 (Redundancy of Duality)

Let $\mathcal{C} \in \mathbf{Cat}_{(\infty,3)}$ and suppose we have an adjunction $L\colon\ + \overset{\perp}{\rightleftarrows} -\ : R$ such that the units and counits ϵ, η admit both left and right adjoints. Then ϵ and η are fully-adjoint.

Proof: Let $\epsilon^r \dashv \epsilon \dashv \epsilon^l$ and $\eta^r \dashv \eta \dashv \eta^l$. By taking left adjoints to each triangle iodentities, we can see that ϵ^l and η^l are respectively a unit and a counit for $R \dashv L$. Similarly, by taking right adjoints to each triangle identities, we can see that ϵ^r and η^r are respectively a unit and a counit for the same adjunction $R \dashv L$. By uniqueness of adjoints, we get $\epsilon^l \simeq \epsilon^r$ and $\eta^l \simeq \eta^r$, which then implies the result. ■

We can now prove our main Theorem.

Theorem 4.2 (Full-Dualizability Criterion)

Let $\mathcal{V}\colon \mathbf{Alg}_{\mathcal{E}_\infty}(\mathbf{Cat}_{(\infty,1)})$ such that its tensor product commutes with sifted colimits in both variables. Let $0 \leq k < n$ and M be a k-morphism in $\mathbf{Mor}_n(\mathcal{V})$. We let A (resp. B) be the (co)domain of M if $1 \leq k$, or otherwise be the identity $\mathbb{1}_{\mathcal{V}}$ if $k = 0$. Then M is fully-dualizable if and only if, for all $0 \leq i \leq n-k$, the

canonical $\mathcal{E}_1$-actions

$$\int_{(S^{i-1},D^i)} (M,A) \curvearrowright M \curvearrowleft \int_{(S^{i-1},D^i)} (M,B) \tag{4.1}$$

induced by the homological maps (see subsection 3.1) $\mathbb{R}^n_k \hookleftarrow \{0\}^{k-1} \times \mathbb{R}_{\geq 0} \times \mathbb{R}^i \times \{0\}^{n-k-i} \twoheadrightarrow \mathbb{R}_{\geq 0}$ and $\mathbb{R}^n_k \hookleftarrow \{0\}^{k-1} \times \mathbb{R}_{\leq 0} \times \mathbb{R}^i \times \{0\}^{n-k-i} \twoheadrightarrow \mathbb{R}_{\geq 0}$ where the second map is the radius (resp. with $\leq$) (see **??** *for details on this construction), both exhibit M as a dualizable module.*

Recall that by convention, we let $S^{-1} = \emptyset$. Note that when $k = 0$, we recover the usual criterion first stated by Lurie [Lur09b, Rem. 4.1.27]:

$$\forall 0 \leq i \leq n, \qquad M \text{ is a dualizable } \int_{S^{i-1}} M\text{-module.}$$

For general k, one can illustrate these $\mathcal{E}_1$-actions as follows:

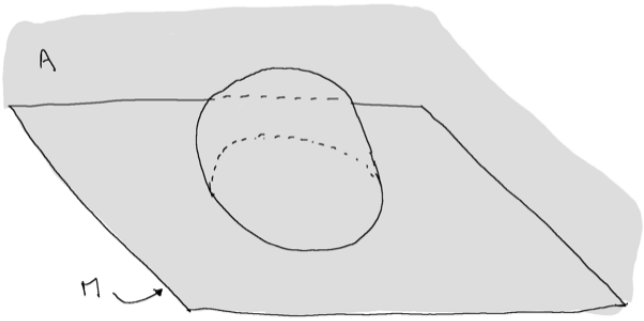

Figure 3: Visualization of the $\mathcal{E}_1$-action on the left-hand side of (4.1) for $i = 2$. The background space is $\mathbb{R}^2 \times \mathbb{R}_{\geq 0}$ with A supported on the 3-dimensional strata (hence seen as a $\mathcal{E}_3$-algebra) and M supported on the 2-dimensional strata (hence seen as a $\mathcal{E}_3$-algebra acted on in a $\mathcal{E}_1$-one-sided-way by A). The $\mathcal{E}_1$-algebra of interest is then the factorization homology of an upper-half dome, with the $\mathcal{E}_1$-structure given by the radial symmetry, and it acts on M pictured at the center of the bottom plane. This factorization homology can be interpreted as a relative form of Hochschild Homology.

Proof: Let $d = n - k$. First, note that the actions described in the statement correspond precisely to all the actions involved in Proposition 3.11 applied to all of the $\sigma(M)$ and $\sigma(M^\circ)$ for $\sigma \in \{\epsilon, \eta\}^d$ a d-duality data. Indeed, one has a commutative diagram of homological maps

$$\begin{tikzcd}
\mathbb{R}^n_k \arrow[d, "(i)"] & \mathbb{R}_{\geq 0} \times \mathbb{R}^i \arrow[l, hook'] \arrow[rr, two heads, "\mathrm{r}"] \arrow[d, hook, "(ii)"] & & \mathbb{R}_{\geq 0} \arrow[d, hook] \\
\mathbb{R}^n_k \arrow[r, "\sim"] & \mathbb{R}^n_{k,\sigma} & \mathbb{R}^n_{k,\overline{\sigma}} \arrow[l, dashed] \arrow[r, two heads] & \mathbb{R}^n_{\overline{n}}
\end{tikzcd}$$

where $a = |\{\sigma = \eta\}|$, (ii) is the inclusion of the coordinates $\{\sigma = \eta\} \subset \{k, \ldots, k + d - 1\}$ and (i) is the obvious permutation of coordinates. One can pick (ii) such that the sign of the permutation is 1, showing that (i) is isotopic to the identity, which in turn exhibits the equivalence claimed between the module structures. Hence, it suffices to show by downward induction on k the following result: M is fully-dualizable if and only if, for all $\sigma \in \{\epsilon, \eta\}^d$ a d-duality data, $\sigma(M)$ and $\sigma(M^\circ)$ have both left and right adjoint, when seen as n-morphisms in $\mathbf{Mor}_n(\mathcal{V})$.

- **Case $k = n - 1$:** By Lemma 4.1, the condition above is equivalent to having both left and right adjoints for each of the units and counits of the full-$\{n-1\}$-adjunction. By Corollary 3.9, these units and counits are described by the algebras $\eta(M)$, $\epsilon(M)$, $\eta(M^\circ)$, $\epsilon(M^\circ)$. By Proposition 3.11, this is equivalent to the condition in the statement of the theorem.
- **Inductive Step:** Note first that by Corollary 2.14 and Corollary 2.16, because $\mathbf{Mor}_n(\mathcal{V})$ has $\{1, \ldots, n-1\}$-adjoints, M is fully-dualizable if and only if we have a lift of the dashed arrow, if and only if we have a lift

of each of the dotted arrows:

$$\begin{tikzcd}
 & \mathsf{Sub}^{\{k+1,\dots,n\}}(\mathbf{Mor}_n(\mathcal{V})) \arrow[r, hook] & \mathbf{Mor}_n(\mathcal{V}) \\
c_{k+1} \arrow[ur, dotted] \arrow[dr, "\epsilon(f^{\vee n})"'] & & \\
 & \mathsf{Free}^{\{k\}}(c_k) \arrow[uu, dashed] & c_k \arrow[l, two heads] \arrow[uu, "M"'] \\
c_{k+1} \arrow[uur, dotted] \arrow[ur, "\eta(f^{\vee n})"'] & &
\end{tikzcd}$$

In other words, M is fully dualizable if and only if all the units and counits involved in the full-$\{k\}$-adjunction are themselves fully-dualizable. As above, by Corollary 3.9, these units and counits are described by the algebras $\eta(M)$, $\epsilon(M)$, $\eta(M^\circ)$, $\epsilon(M^\circ)$. By the inductive hypothesis, duality is then equivalent to: for all $\sigma \in \{\epsilon, \eta\}^{d-1}$ a $(d-1)$-duality data, $\sigma(\eta(M))$, $\sigma(\epsilon(M))$, $\sigma(\eta(M^\circ))$, $\sigma(\epsilon(M^\circ))$, $\sigma(\eta(M)^\circ)$, $\sigma(\epsilon(M)^\circ)$, $\sigma(\eta(M^\circ)^\circ)$ and $\sigma(\epsilon(M^\circ)^\circ)$ all have left and right adjoint, when seen as n-morphisms in $\mathbf{Mor}_n(\mathcal{V})$. Using Lemma 3.10 and the exchange property of $\circ$ with $\eta \leftrightarrow \epsilon$, we can conclude the proof. ■

In particular, dualizable module $\mathcal{E}_1$-actions are well-understood in the category $\mathcal{S}p$ of spectra and we get this immidiate corrolary.

Corollary 4.3 (Full-Dualizability of Ring Spectra)

Let $0 \leq k < n$ and M be a k-morphism in $\mathbf{Mor}_n(\mathcal{V})$. Then M is fully-dualizable if and only if, for all $1 \leq i \leq n-k$, the canonical $\mathcal{E}_1$-actions

$$\int_{(S^{i-1},D^i)} (M, A) \curvearrowright M \curvearrowleft \int_{(S^{i-1},D^i)} (M, B)$$

induced by the homological maps (see subsection 3.1) $\mathbb{R}^n_{\frac{}{k}} \hookleftarrow \{0\}^{k-1} \times \mathbb{R}_{\geq 0} \times \mathbb{R}^i \times \{0\}^{n-k-i} \twoheadrightarrow \mathbb{R}_{\geq 0}$ where the second map is the radius (resp. with $\leq$), both exhibit M as a perfect/compact module.

Proof: This is a direct consequence of Theorem 4.2 and [Lur17, Prop. 7.2.4.4]. ■

Remark 4.1 (Characterizing Full-Dualizability in Other Higher Categories). Implicitly, the proof uses that: if f is a fully-$\{k, \dots, l\}$-dualizable k-morphism for $k \leq l$, full-$\{k, \dots, l+1\}$-dualizability is equivalent to all $\sigma(f)$ having left adjoints for $\sigma \in \{\epsilon, \eta\}^{l-k+1}$ a $(l-k+1)$-duality data. This can be proved independently in any higher category with the tools of section 2 following the same argument. It would simply remain to show a few more "redundancy" lemmas. We avoid this argument here because, in our case, the Morita category is built with a natural symmetry $\mathbf{Aut}(X) \curvearrowright \mathbf{Alg}_{/X}$, which is explicit with factorization homology (see Definition 3.2).

Remark 4.2 (Other Proof Strategies). There are many possible proof strategies for this result, from closest to farthest from the one presented here:

- One could do a upward induction on k instead, using some right associativity lemma for the pinch constructions. This lemma is significantly heavier to spell out as the stratifications involved are more complicated, but the core idea is the same.
- One could also imagine a more elementary proof of the result above without explicit description of the pinch maps, by simply writing naïvely an inductive argument. This would however not translate well to the proof of Theorem B, where the explicit description of the pinch maps is very helpful to elucidate Proposition 3.15.
- One could also try to write the (∞, n)-functor $\mathcal{B}\mathrm{ord}^{\mathrm{fr}}_n \to \mathbf{Mor}_n(\mathcal{V})$ and use handlebody decomposition to identify which actions on the top n-morphism level need to be dualizable. This is a much more technical approach. Such a functor was constructed in [Sch14] for a pointed version of $\mathbf{Mor}_n(\mathcal{V})$ with the language of factorization algebras and relative categories. A proof following this approach would involve re-proving many technical results already present in [AFT17b, AFR18a, AFT17a] under a different formalism, or showing an equivalence of both formalisms. We avoid this by usng a crucial idea: deciding if a morphism is fully-dualizable is not necessarily a homotopy-coherent problem.

4.2 Theorem B

Similarly to the proof of Theorem A, we start with a redundancy lemma. Only this time, we are interested in the redundancy of invertibility properties.

Lemma 4.4 (Redundancy of Invertibility)

Let $\mathcal{C} \in \mathbf{Cat}_{(\infty,2)}$ and suppose we have two parallel adjunctions

$$\begin{tikzcd} \bullet \arrow[r, "f"] \arrow[r, bend left=60, "\perp"] & \bullet \arrow[l, bend left=60, "\perp"] \end{tikzcd}$$

in $\mathcal{C}$ such that the units are each time equivalences. Then f is an equivalence.

Proof: Note that equivalences and adjoints are detected in the homotopy 2-category. Without loss of generality, one can then assume that $\mathcal{C}$ is a 2-category. In this case, one can use the 2-categorical Yoneda lemma [JY21] to reduce to the case where $\mathcal{C} = \mathbf{Cat}_{(\infty,1)}$. In $\mathbf{Cat}_{(1,1)}$, the unit of the bottom adjunction being a natural isomorphism implies that f is fully faithful, which in turn implies that the counit of the top adjunction is a natural isomorphism. Hence, the top adjunction exhibits f as an (adjoint) equivalence. ■

Using the above lemma, we can then prove our second criterion.

Theorem 4.5 (Invertibility Criterion)

Let $0 \leq k < n$ and $\mathcal{V} \in \mathbf{Alg}_{\mathcal{E}_\infty}(\mathbf{Cat}_{(\infty,1)})$. Let M be a k-morphism in $\mathbf{Mor}_n(\mathcal{V})$. Then M is invertible if and only if it is fully-dualizable and, for all $1 \leq i \leq n-k$, the universal $\mathcal{E}_{n-k-i+1}$-algebra maps

$$\int_{(S^{i-1},D^i)} (M, A) \xrightarrow{\sim} \mathbf{HC}^B_{\mathcal{E}_{n-k-i}}(M)$$
$$\int_{(S^{i-1},D^i)} (M, B) \xrightarrow{\sim} \mathbf{HC}^A_{\mathcal{E}_{n-k-i}}(M)$$

defined in Proposition 3.15 are equivalences.

Fundamentally, one can interpret invertibility of these maps as saying: $(i-1)^{\text{th}}$-higher Hochschild Homology of M relative to A computes the universal $\mathcal{E}_{k-n-i+1}$-algebra acting on M in a B-linear way (and vice-versa). Note that once again, we recover Lurie's criterion for $k = 0$, and more familiarly, we recover the usual conditions for an Azumaya algebras when $n = 1$ and $k = 0$: M is dualizable and

$$k \xrightarrow{\sim} \mathbf{HC}_{\mathcal{E}_1}(M)$$
$$M^{\text{op}} \otimes_k M \xrightarrow{\sim} \mathbf{End}_k(M)$$

Proof: By Lemma 2.18, the functor $\mathsf{Free}^{\{k,\ldots,n\}}(c_k) \twoheadrightarrow c_{k-1}$ is epimorphic and correspond under precomposition to some $[c_k, \mathbf{Mor}_n(\mathcal{V})]_{\simeq} \hookrightarrow [\mathsf{Free}^{\{k,\ldots,n\}}(c_k), \mathbf{Mor}_n(\mathcal{V})]$. Hence, any invertible k-morphism is automatically fully-dualizable. We can then assume M is fully-dualizable, and show invertibility is equivalent to all these maps of algebra being invertible.

Let $d = n - k$ (which can be zero). Similarly to Theorem A, now using Proposition 3.15, we first observe that invertibility of all these maps is precisely equivalent to the statement: for all $\sigma \in \{\epsilon, \eta\}^d$ a d-duality data, the unit of $\sigma(M) \dashv \sigma(M)^\vee$ and of $\sigma(M)^\circ \dashv (\sigma(M)^\circ)^\vee$ are invertible. As a consequence, it suffices to show by downward induction on k the equivalence between invertibility of M and this condition.

- **Base Case $k = n$:** this is an immediate of consequence of the lemma above.
- **Inductive Step:** By Lemma 2.17 and Corollary 3.9, M is invertible if and only if $\epsilon(M)$ and $\eta(M)$ are invertible as $(k+1)$-morphisms in $\mathbf{Mor}_n(\mathcal{V})$. The statement then follows from the inductive hypothesis and Lemma 3.10. ■

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